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CORNGUARD: A Mobile Application for Real-Time Corn Disease Detection Using Convolutional Neural Networks

Course Code and Course Title

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TABLE OF CONTENTS

CHAPTER I.....	4
1.1 Background of the Study.....	4
1.2 Statement of the Problem.....	8
1.3 Objectives of the Study.....	9
1.3.1 General Objective.....	9
1.3.2 Specific Objectives.....	10
1.4 Significance of the Study.....	12
1.5 Scope and Limitations.....	14
1.5.1 Scope.....	14
1.5.2 Limitations.....	16
1.5.3 Implementation of the Community Discussion Platform.....	17
1.6 Comparison of Existing Systems.....	19
1.7 Definition of Terms.....	21
Technical & System Terms.....	21
Agricultural & Biological Terms.....	22
CHAPTER II.....	24
2.0 Review of Related Studies.....	24
Deep Learning and CNNs for Plant Disease Detection.....	24
MobileNetV2: Lightweight Architecture for Mobile Deployment.....	28
Mobile Applications for Agricultural Disease Management.....	29
Community-Based Platforms and Peer-to-Peer Knowledge Sharing for Agricultural Disease Management.....	30
AI Accessibility for Smallholder Farmers in Developing Regions.....	32
TensorFlow Lite and On-Device Offline Inference.....	34
CHAPTER III.....	36
3.0 Development Model.....	36
3.1 Phase 1: Planning and Requirement Analysis.....	37
3.1.1 Requirements Gathering.....	37
3.1.2 Dataset Collection and Image Acquisition.....	38
3.1.3 Data Preprocessing.....	40
3.2 Feasibility Study.....	42
3.2.1 Technical Feasibility.....	42
3.2.2 Operational Feasibility.....	42
3.2.3 Economic Feasibility.....	42

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

3.3 Defining Requirements (Software Requirements Specification).....	43
3.3.1 Functional Requirements.....	43
3.3.2 Non-Functional Requirements.....	44
3.4 Phase 2: System Design.....	44
3.4.1 System Architecture.....	44
3.4.2 Data Flow Diagrams (Level 0 and Level 1).....	46
3.4.3 Use Case Diagram.....	48
3.4.4 Database Schema.....	49
3.4.5 Algorithm Used.....	50
3.4.6 Hardware Requirements.....	55
3.4.7 Software Stack.....	56
3.4.8 Software Requirements.....	58
3.5 Phase 3: Testing.....	60
3.5.1 Model Evaluation.....	60
1. Accuracy.....	61
2. Precision.....	61
3. Recall.....	61
4. F1-Score.....	62
5. Confusion Matrix.....	62
6. Validation and Testing Procedure.....	62
7. TensorFlow Lite Performance Evaluation.....	62
3.5.2 System Evaluation and Testing.....	63
3.5.3 User Acceptance Testing (UAT).....	67
3.5.4 User Interface and Experience Design.....	68
3.6 Project Timeline.....	71
References:.....	72

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Corn (*Zea mays*) is one of the most widely cultivated crops in the world and plays a vital role in global food security, directly supporting the objectives of Sustainable Development Goal 2 (Zero Hunger), which aims to end hunger, achieve food security, and promote sustainable agriculture (United Nations, 2015). As a staple crop, corn serves as a primary source of nutrition for humans and as a key component in animal feed, making it essential to both agricultural production and economic stability in many countries (FAO, 2022).

However, corn production is significantly threatened by various plant diseases such as Common Rust, Northern Leaf Blight, and Gray Leaf Spot. These diseases can lead to substantial reductions in crop yield and quality, ultimately affecting food supply chains and farmer livelihoods. Studies show that disease outbreaks can cause yield losses ranging from 15% to 50% if not properly managed (Khan et al., 2023; Zhu et al., 2024). Such losses pose a serious challenge to achieving global food security goals, especially in developing regions where agriculture is a primary source of income.

Early and accurate detection of corn diseases is therefore critical in minimizing crop damage and ensuring sustainable agricultural productivity. Traditionally, disease identification relies on manual inspection by farmers or agricultural experts. While this method has been widely practiced, it is often time-consuming, subjective, and prone to human error, particularly in large-scale farms or remote areas with limited access to experts (Barbedo, 2018).

With the rapid advancement of artificial intelligence and computer vision technologies, automated plant disease detection systems have emerged as a promising solution. In particular, Convolutional Neural Networks (CNNs) have demonstrated high accuracy in classifying plant diseases from leaf images, offering a faster and more reliable alternative to traditional methods

(Mohanty et al., 2016; Too et al., 2019). These technologies have the potential to significantly improve disease management practices and contribute to global efforts in achieving food security under SDG 2.

In the Philippines, corn is one of the most important agricultural commodities, serving as a major source of food for human consumption and as a primary component in livestock and poultry feed. The country's agricultural sector heavily depends on corn production, particularly in rural areas where farming is a primary source of income. However, the tropical climate of the Philippines, characterized by high humidity and frequent rainfall, creates favorable conditions for the development and spread of plant diseases such as Common Rust, Northern Leaf Blight, and Gray Leaf Spot. These diseases significantly affect crop productivity and can lead to substantial economic losses for farmers (Barbedo, 2018; Too et al., 2019).

Despite the importance of corn production, many Filipino farmers still rely on traditional methods of disease identification, which involve visual inspection and personal experience. This approach is often unreliable, especially when symptoms of different diseases appear similar, leading to misdiagnosis and inappropriate treatment. Furthermore, access to agricultural experts and plant pathologists is limited in many rural areas, making it difficult for farmers to obtain timely and accurate diagnoses (Khan et al., 2023).

Recent studies have shown that deep learning models can effectively classify plant diseases from leaf images with high accuracy, providing a more reliable alternative to manual inspection (Mohanty et al., 2016; Too et al., 2019). However, the adoption of such technologies in the Philippines remains limited due to factors such as lack of awareness, insufficient access to digital tools, and dependence on internet connectivity. These challenges highlight the need for a localized, accessible, and user-friendly solution that can support Filipino farmers in improving crop management practices and increasing agricultural productivity.

Bukidnon is recognized as one of the key agricultural regions in the Philippines, where corn production plays a significant role in supporting both the local food supply and the livestock industry. The region contributes a substantial portion of the country's total corn output, making it

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

a vital area for agricultural development and food security. However, corn farmers in Bukidnon continue to face persistent challenges related to plant diseases such as Northern Leaf Blight, Gray Leaf Spot, and Common Rust, which thrive under the region's warm and humid environmental conditions. These diseases can spread rapidly across farms, especially when early detection and proper management are not implemented.

Despite the importance of corn farming in the region, many farmers still rely on traditional and experience-based methods for disease identification. This practice is often subjective and prone to error, particularly when disease symptoms overlap or are not easily distinguishable. In addition, access to agricultural experts and diagnostic services remains limited in remote farming communities, resulting in delayed or incorrect treatment decisions (Khan et al., 2023). Such limitations can lead to significant yield losses and increased production costs due to ineffective pesticide use.

According to the Philippine Statistics Authority, Bukidnon contributed 68.1% of Region X's total corn production of 1,520,889 metric tons in 2023 — the highest provincial share in the region — and ranked first in the country in total agricultural and fisheries production value at ₱155.28 billion (PSA, 2023). Given this scale of production, even a small percentage of disease incidence can result in considerable economic losses for the province. Studies suggest that corn diseases can reduce yield by 15% to 50% depending on severity and management practices (Zhu et al., 2024). Without accurate and timely disease detection, farmers may experience reduced harvest quality and income instability.

The integration of mobile-based artificial intelligence systems presents a promising solution to these challenges. By enabling real-time and accessible disease detection through smartphone devices, such technologies can empower farmers to identify infections at an early stage and apply appropriate interventions. This approach not only improves crop productivity but also supports sustainable agricultural practices in the region.

At the local level, a significant number of corn farmers in Bukidnon continue to rely on personal experience, advice from neighboring farmers, or traditional practices to identify crop

diseases. This approach is often subjective and prone to error, especially when unfamiliar symptoms appear on corn leaves. Such misdiagnoses frequently lead to delayed interventions, improper application of pesticides, and unnecessary financial losses. In some cases, incorrect treatment not only fails to resolve the disease but can also damage the crop further or negatively impact the environment through excessive chemical use (Khan et al., 2023; Barbedo, 2018).

Interviews with local farmers have revealed that many lack access to reliable diagnostic tools and are unfamiliar with emerging technologies that could aid in disease identification. Farmers often face challenges such as limited internet connectivity, outdated or low-end mobile devices, and insufficient knowledge of optimal treatment methods. These barriers hinder the timely and accurate detection of corn diseases, resulting in preventable crop losses and reduced productivity.

To address these challenges, this study proposes the development of CORNGUARD, a mobile application designed to provide real-time, accurate, and offline detection of corn leaf diseases. By leveraging a lightweight deep learning model, MobileNetV2, the application can analyze leaf images directly on the user's device, allowing farmers to obtain instant diagnosis and recommendations without relying on internet connectivity. The system is tailored to local conditions, considering factors such as lighting, leaf variations, and common disease patterns observed in Bukidnon farms.

The proposed application empowers farmers by providing a practical, accessible, and reliable tool for early disease detection. By integrating AI technology with localized agricultural knowledge, CORNGUARD aims to improve crop management, reduce financial losses, and support sustainable farming practices in the region. Furthermore, the system has the potential to generate valuable data on disease incidence, which can assist local agricultural offices in monitoring trends and making informed decisions regarding resource distribution and preventive strategies.

1.2 Statement of the Problem

Corn diseases remain a major challenge for farmers in Bukidnon and across the Philippines because traditional methods of disease identification rely heavily on manual observation and expert knowledge. Many farmers, particularly those in remote or rural areas, do not have immediate access to agricultural specialists who can accurately diagnose plant diseases (Barbedo, 2018; Khan et al., 2023). This often leads to misdiagnosis, delayed treatment, and the inappropriate use of pesticides, which can result in reduced crop yields, financial losses, and negative environmental impacts.

Although artificial intelligence-based plant disease detection systems have been developed, many existing solutions are not well-suited for real-world agricultural conditions in rural communities. Most systems require stable internet connectivity and high-performance devices, making them less accessible to smallholder farmers. Additionally, these models are often trained using controlled datasets that do not account for real-world variability such as inconsistent lighting, complex backgrounds, and variations in leaf appearance (Mohanty et al., 2016; Too et al., 2019). Consequently, their effectiveness and reliability may be significantly reduced when applied in actual field environments.

Moreover, there is a lack of localized and accessible mobile applications specifically designed for corn disease detection in the Philippine context. Existing tools often fail to consider regional agricultural conditions and the technological limitations of farmers in developing regions. This highlights a critical gap in providing a practical, lightweight, and offline-capable solution that can deliver accurate and real-time disease detection directly in the field.

Given these challenges, there is a need to develop a mobile-based system that integrates deep learning techniques for accurate and efficient corn disease detection. Therefore, this study proposes the development of CORNGUARD, a mobile application designed to detect corn leaf diseases using image-based analysis. The system aims to operate offline, be optimized for low-end mobile devices, and be tailored to local agricultural conditions in Bukidnon.

Specifically, the study seeks to answer the following research questions:

1. What deep learning model can be implemented to detect corn diseases from leaf images?
2. How can a mobile application be developed to integrate the deep learning model for real-time disease detection?
3. How accurate is the developed model in classifying different corn leaf diseases?
4. What performance metrics (accuracy, precision, recall, and F1-score) can be achieved by the proposed system?

1.3 Objectives of the Study

1.3.1 General Objective

To develop CORNGUARD, an AI-powered mobile application for automated corn disease detection, GIS-based outbreak monitoring, heatmap visualization, and location-based notification services for smallholder corn farmers in Bukidnon. The system will use the MobileNetV2 deep learning model to classify Common Rust, Gray Leaf Spot, Northern Leaf Blight, and Healthy corn leaves. MobileNetV2 was selected due to its high accuracy, lightweight architecture, and efficient deployment on low-end Android devices using TensorFlow Lite, enabling offline and real-time disease detection in rural farming areas with limited internet connectivity. The system will also integrate GIS technology to display the exact location, barangay, municipality/city, and province of detected disease cases, generate heatmaps based on the outbreak, and automatically notify nearby farmers of severe disease outbreaks. Furthermore, the model provides a future upgrade path toward more advanced lightweight architectures such as EfficientNet-Lite0 while remaining suitable for resource-constrained mobile devices.

1.3.2 Specific Objectives

1. To collect and preprocess a dataset of corn leaf images containing healthy and diseased samples:
 - Common Rust



Figure 1. Corn leaf infected with Common Rust (Puccinia sorghi).

- Gray Leaf Spot



Figure 2. Corn leaf specimen displaying rectangular, grey-to-tan lesions characteristic of Gray Leaf Spot (Cercospora zeae-maydis).

- Northern Leaf Blight



Figure 3. Corn leaf specimen exhibiting elongated, elliptical "cigar-shaped" lesions indicative of Northern Leaf Blight (Exserohilum turcicum).

- Healthy leaves



Figure 4. Healthy corn leaf (Zea mays) specimen used as a baseline control, showing uniform green pigmentation and no visible signs of infection.

2. To evaluate and train a MobileNetV2-based Convolutional Neural Network (CNN) model capable of accurately classifying corn leaf diseases using TensorFlow and TensorFlow Lite.

3. To integrate the trained TensorFlow Lite model into an Android mobile application capable of performing offline and real-time corn disease detection.
4. To detect and classify three target maize diseases, namely Gray Leaf Spot, Northern Leaf Blight, and Common Rust.
5. To evaluate the performance of the model using metrics such as accuracy, precision, recall, and F1-score.
6. To develop a GIS-based disease monitoring system capable of identifying and displaying the exact location, barangay, municipality/city, and province of reported corn disease cases in Bukidnon.
7. To implement a heatmap visualization system that identifies and highlights areas with high concentrations of severe corn disease outbreaks to provide early warning information to nearby farmers.
8. To implement a community notification and reporting module that allows farmers to share disease-related information, outbreak updates, and treatment recommendations.

1.4 Significance of the Study

The development of CORNGUARD, a mobile-based disease detection system for corn leaves using MobileNetV2, provides substantial benefits to multiple stakeholders and aligns directly with the study's objectives and features.

1. Farmers: Local small-scale and commercial corn farmers are the primary beneficiaries. The application offers a portable, offline-capable diagnostic tool for early detection of diseases such as Common Rust, Gray Leaf Spot, and Northern Leaf Blight. By providing faster and more accurate disease identification, farmers can apply appropriate treatments promptly, reduce crop losses, and improve productivity. Furthermore, the integrated community platform empowers farmers to communicate with one another about active disease occurrences in their area — enabling peer-to-peer outbreak warnings, treatment discussions, and shared field knowledge. Research

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

demonstrates that social networks and interactions among farmers significantly influence the adoption of agricultural innovation, particularly among smallholder communities in developing regions (An & Mahasuweerachai, 2025). Studies on technology integration among corn farmers in Bukidnon confirm that limited digital literacy and insufficient access to modern tools remain key barriers to agricultural productivity improvement in the province (Corroz et al., 2025). Given that Bukidnon accounted for 68.1% of Region X's total corn production in 2023 and ranked first nationally in agricultural production value at ₱155.28 billion (PSA, 2023), even modest improvements in disease detection and treatment timing can yield significant economic benefits for the province.

2. Agricultural Technicians and Extension Workers: CORNGUARD serves as a digital assistant for field workers, automating initial disease diagnosis. This allows technicians to cover larger areas, focus on complex cases, and efficiently allocate resources for optimal disease management.

3. Local Government Units (LGUs) and Agricultural Offices: The application's disease history tracking module enables the generation of data for barangay-level disease incidence mapping in Bukidnon. This supports local agricultural offices, such as the CDO City Agriculture Office, in monitoring emerging outbreaks and strategically allocating resources or pesticides (BukSU, 2025). Additionally, community-reported disease posts from the platform's social forum can serve as a crowdsourced early warning layer, supplementing official monitoring with real-time, ground-level observations from farmers themselves — a model validated by Plantix's geo-time-tagged community upload system, which enabled live tracking of pest and disease outbreaks across India and Southeast Asia (CGIAR, 2020). CORNGUARD thus fills a critical data gap currently unaddressed in the region.

4. Environment: Accurate classification and targeted treatment recommendations help prevent unnecessary or excessive pesticide use. By implementing precise interventions, farmers can reduce environmental impact, promoting sustainable farming practices.

5. Future Researchers: This study contributes to the growing body of research on deep learning applications in agriculture, particularly in mobile-based plant disease detection. The findings provide a foundation for future studies, including:

- MobileNetV2-based corn disease classification validated against real Bukidnon field conditions.
- Evaluation of community-based disease reporting platforms integrated within AI diagnostic tools — examining information credibility, adoption rates, and the effectiveness of peer-to-peer outbreak warnings in smallholder farming communities in the Philippines (Aquino & Concepcion, 2025).

1.5 Scope and Limitations

1.5.1 Scope

The study focuses on the development of a mobile-based diagnostic tool utilizing a Deep Learning architecture, specifically MobileNetV2, for identifying common corn leaf diseases. The system combines a deep learning component (MobileNetV2 via transfer learning) with a knowledge-based component (rule-based agronomic modules). The system is designed to assist farmers by providing real-time, image-based disease detection through a smartphone application.

1. **Target Crop and Diseases:** The system will identify Common Rust, Gray Leaf Spot, and Northern Leaf Blight, along with healthy leaf samples for comparison and classification.
2. **Technology Stack:** The system uses a MobileNetV2 deep learning architecture, which is designed to be lightweight and optimized for mobile devices. The model is trained using transfer learning, where a pre-trained MobileNetV2 model trained on the ImageNet dataset is adapted to classify corn leaf diseases.

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

3. **Core Functionality:** The application will include real-time image capture and offline inference, allowing users to diagnose corn leaf diseases without requiring an internet connection. The application integrates two core modules: (a) offline CNN inference via TensorFlow Lite; and (b) heatmap grading. The system analyzes images of corn leaves using a trained deep learning model and provides disease classification results directly on the mobile device. In addition, the application will provide general treatment recommendations and preventive measures based on agricultural references to assist farmers in managing detected diseases effectively.
4. **Data Source:** Training will be conducted using a merged dataset of corn leaf images sourced from public repositories such as Kaggle and the PlantVillage dataset developed by Pennsylvania State University. Locally captured corn leaf images from Bukidnon will be collected to improve regional relevance and model generalization.
5. **Farmer Validation Component:** The study will include interviews with selected local corn farmers to understand current disease identification practices, pesticide usage, and challenges encountered in the field. This will ensure that the proposed system aligns with real agricultural needs.
6. **Community Discussion Platform:** The application will include an internet-connected community forum module that allows farmers to post disease reports with optional photo attachments, share treatment outcomes, warn nearby farmers of emerging outbreaks, and participate in topic-based discussions about corn health. Posts are organized by disease type and location to facilitate localized, actionable communication. This module is inspired by the community feature model validated in Plantix, where geo-time-tagged uploads and push notifications enabled real-time tracking of pest and disease movements across farming regions (CGIAR, 2020; PEAT/GSMA, 2025). The community module requires internet connectivity and is designed to complement — not replace — the offline CNN inference module.

1.5.2 Limitations

Although data augmentation techniques will be applied to simulate varying lighting conditions, extreme weather conditions such as heavy rain, excessive glare, or low-light environments may affect image quality and diagnostic accuracy.

1. **Image Specificity:** The system is strictly limited to leaf-based diseases of corn. It does not diagnose diseases affecting roots, stems, or internal plant structures that do not visibly manifest on the leaves.
2. **Disease Coverage Limitation:** The model will only classify the selected diseases included in the training dataset. It may not accurately identify rare or newly emerging corn diseases outside the defined categories.
3. **Hardware Dependency:** The speed of diagnosis (inference time) depends on the smartphone's processing capability. Older devices with limited RAM or lower processing power may experience delays.
4. **Treatment Scope:** The application provides general guidance on possible treatments but does not replace professional consultation with licensed agriculturists or laboratory-based diagnosis.
5. **Community Platform Content Reliability:** The community discussion module relies on user-generated content, which may include inaccurate disease identifications, anecdotal treatment advice, or unverified outbreak reports. As noted by Aquino and Concepcion (2025) in their study of Filipino farmers using social media for crop disease information, credibility of information shared on digital platforms is a recognized challenge among rural agricultural communities. The system does not include automated fact-checking or expert moderation, and farmers are advised to use community posts as supplementary guidance rather than a replacement for verified agricultural expertise.

1.5.3 Implementation of the Community Discussion Platform

The community platform is implemented as an online module embedded within the CORNGUARD mobile application, designed to be accessible whenever the user's device has internet connectivity. The following describes the key technical and functional components of the implementation:

a) Post and Thread Structure. The forum adopts a Reddit-style threaded discussion model, where each post contains a title, a body description, an optional photo attachment (typically a photo of the diseased corn leaf taken directly within the app), a disease tag (e.g., Common Rust, Northern Leaf Blight, Gray Leaf Spot, or Unknown), a timestamp, and the poster's barangay or municipality. Farmers can reply to posts, enabling multi-turn discussions. Posts can be upvoted by other users to surface the most helpful reports within the community feed. This structure is consistent with the community feature architecture validated in Plantix, which similarly allows farmers to help one another diagnose plant diseases through threaded discussions (CGIAR, 2020; PEAT/GSMA, 2025).

b) Integration with CNN Detection Results. When a farmer completes a disease scan using the offline CNN module, CORNGUARD provides an option to automatically pre-populate a community post with the detected disease type, the captured leaf image, and the device's approximate location. This reduces the input burden on farmers and ensures that community reports are anchored to verified AI diagnostic results rather than relying solely on the user's own disease identification. This design mirrors the geo-time-tagging workflow used in Plantix, where every uploaded image was tagged with location data to build a spatial map of disease incidence across regions (CGIAR, 2020).

c) Backend and Data Storage. The community module is implemented using a cloud-based backend — specifically Firebase Realtime Database or Firestore — to store posts, replies, and user data. Firebase was selected for its offline-to-online synchronization capability, low implementation cost, and suitability for Android

development environments. Post data is stored as structured JSON documents containing post ID, user ID, disease tag, location metadata, image URL, post body, timestamp, and reply thread. Images uploaded to posts are stored in Firebase Cloud Storage. This lightweight, serverless backend approach is consistent with the architecture commonly adopted in community-based agri-tech mobile applications (Rao et al., 2025).

d) Location-Based Feed and Outbreak Filtering. The community feed is filtered by geographic proximity — displaying posts from the same barangay, municipality, or province first — to ensure that outbreak warnings are contextually relevant to the farmer viewing the feed. Farmers can also filter the feed by disease type to quickly find discussions relevant to a specific pathogen affecting their crops. This location-aware architecture is directly informed by Plantix's geo-time-tagged community system, which demonstrated that spatially filtered disease reports enable faster community-level outbreak recognition and response (CGIAR, 2020; PEAT/GSMA, 2025).

e) User Authentication and Roles. Farmers register using a mobile number or email address. Upon registration, users provide their farm's barangay and municipality to enable location-based feed filtering. The system implements two user roles: (1) Farmer — can post, reply, and upvote; and (2) Agricultural Technician — an optional role for registered extension workers who can mark posts as verified or provide expert replies. This role structure is intended to address the credibility challenge identified by Aquino and Concepcion (2025), who found that information reliability was a key concern among Filipino farmers using social media for crop disease information.

f) Notification System. When a farmer in the same municipality posts a new disease report tagged with a specific pathogen, the application sends a push notification to other registered CORNGUARD users in the same area. This outbreak alert mechanism is designed to replicate the push notification system used by Plantix, which notified farmers in a given area whenever nearby users uploaded reports of the same pest or

disease — enabling proactive community-level responses before outbreaks spread across farms (CGIAR, 2020). Push notifications are implemented using Firebase Cloud Messaging (FCM), which is compatible with low-end Android devices and operates over mobile data connections.

g) Connectivity Requirement and Offline Behavior. The community module requires an active internet connection to post, browse, and receive notifications. Users who are offline retain access to previously loaded posts cached on the device. The offline CNN inference module operates independently of the community platform and remains fully functional without connectivity. This hybrid architecture — offline AI detection paired with an online community layer — ensures that the core diagnostic functionality is never compromised by connectivity limitations, while the community features remain available to users who have periodic internet access, which is consistent with the digital infrastructure realities of smallholder farming communities in Bukidnon (Aquino & Concepcion, 2025; Farmonaut, 2025).

1.6 Comparison of Existing Systems

To evaluate the positioning of the proposed CORNGUARD application, a comparative analysis was performed against established benchmarks. On an international level, the Ag PhD Corn Diseases application by IFA Productions (2025) serves as a prominent digital field guide; however, it remains a manual reference tool that requires human interpretation. Locally, a primary reference is the BukSU Corn Disease Detector, an AI-driven tool developed through the research of Lapates (2024) at Bukidnon State University. This system, which utilizes the YOLOv8 architecture, has been formally transitioned into field use through a partnership with the Malaybalay City Agriculture Office (BukSU, 2026). While the BukSU system demonstrates high precision in identifying localized corn pathogens, CORNGUARD differentiates itself by focusing on the MobileNetV2 framework. This choice provides a more lightweight and

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resource-efficient solution, specifically optimized for farmers using entry-level mobile hardware in real-time field conditions.

Features / Criteria	Manual Farmer Inspection	Ag PhD Corn Diseases	BukSU Corn AI	Proposed CORNGUARD
Uses Artificial Intelligence	✗	✗	✓	✓
Detects Corn Leaf Diseases	✓	✓	✓	✓
Real-Time Disease Detection	✗	✗	✓	✓
Works on Mobile Devices	✗	✓	✓	✓
Core Detection Requires Internet	✗	✗	✗	✗
Community Module Requires Internet	✗	✗	✗	✓
Designed Specifically for Corn	✗	✓	✓	✓
High Disease Classification Accuracy	Low (Subjective)	Low (Manual)	High	High
Supports Farmers in Field Conditions	✓	✓	✓	✓
Provides Instant Diagnosis	✗	✗	✓	✓

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

Lightweight AI Model for Mobile Use	✗	✗	✓ (YOLOv8)	✓ (MobileNetV2)
Community Disease Forum / Outbreak Reporting	✗	✗	✗	✓ (CORNGUARD)

1.7 Definition of Terms

Technical & System Terms

1. Artificial Intelligence (AI): A field of computer science that enables machines to perform tasks that normally require human intelligence.
2. Convolutional Neural Network (CNN): A deep learning model specifically designed for image recognition and classification tasks.
3. Deep Learning: A subset of machine learning that utilizes multi-layer neural networks to analyze and extract features from large datasets.
4. Mobile Application: Software designed to run on portable devices such as smartphones or tablets.
5. TensorFlow Lite: A technology used to optimize and integrate deep learning models into mobile devices for real-time classification.
6. Transfer Learning: A process where a pre-trained model (such as MobileNetV2 or VGG16) is adapted to a new, specific task to improve performance and efficiency.
7. Real-Time Disease Detection: The immediate process of identifying plant diseases from leaf images captured through a mobile device's camera.
8. MobileNetV2: A lightweight convolutional neural network architecture developed by Google, designed for mobile and embedded applications. It uses depthwise separable convolutions and inverted residual blocks to achieve competitive accuracy with significantly reduced computational cost compared to standard CNNs.
9. Community Forum / Discussion Platform: An online feature integrated within a mobile application that allows registered users to create posts, share photos, comment, and

exchange information on a specific topic — in this context, corn crop health and disease management. Inspired by platforms such as Reddit and Plantix's community feature, CORNGUARD's forum is organized by disease type and location to facilitate targeted, actionable peer-to-peer communication among corn farmers (Rao et al., 2025; CGIAR, 2020).

10. Crowdsourcing: The practice of collecting information or input from a large group of users, particularly farmers, to generate a collective dataset or knowledge base. In CORNGUARD, community disease reports constitute a form of crowdsourced outbreak surveillance (PEAT/GSMA, 2025).
11. Peer-to-Peer Knowledge Sharing: The horizontal exchange of practical information, experiences, and advice between individuals of similar background and expertise — in this context, between corn farmers — rather than through formal top-down expert advisory channels (An & Mahasuweerachai, 2025).
12. Firebase: A cloud-based mobile and web application development platform by Google, providing real-time database, cloud storage, and push notification services. Used in CORNGUARD to power the community forum's backend data storage and the outbreak notification system.
13. Push Notification: An automated alert sent to a user's mobile device by an application, even when the app is not actively open. In CORNGUARD, push notifications are triggered when nearby farmers post a new disease outbreak report, implemented via Firebase Cloud Messaging (FCM).

Agricultural & Biological Terms

1. Corn (*Zea mays*): A widely cultivated crop that is a major source of food and animal feed.
2. Common Rust (*Puccinia sorghi*): A fungal disease that infects corn leaves, characterized by small, reddish-brown pustules.
3. Gray Leaf Spot (*Cercospora zeae-maydis*): A disease that manifests as rectangular, grey-to-tan lesions on corn leaf specimens.

University of Science and Technology of the Southern Philippines

Alubijid | Cagayan De Oro | Claveria | Jasaan | Oroquieta | Panaon

4. Northern Leaf Blight (*Exserohilum turcicum*): An infection identified by elongated, elliptical "cigar-shaped" lesions on the leaf. Yield losses of 15–50% are documented when infection establishes before the silking (R1) growth stage. (De Rossi et al. (2022), Crop Protection 151).

CHAPTER II

REVIEW RELATED LITERATURE

This chapter presents a synthesis of relevant studies and literature that serve as the theoretical and technical foundation for the development of CORNGUARD — a mobile application for real-time corn disease detection using MobileNetV2. The review is organized thematically to reflect the core components of the proposed system.

2.0 Review of Related Studies

Deep Learning and CNNs for Plant Disease Detection

The application of deep learning, particularly Convolutional Neural Networks (CNNs), to plant disease detection has become a rapidly expanding research area over the past decade. Mohanty et al. (2016) pioneered large-scale benchmarking using the PlantVillage dataset and demonstrated that deep learning models could achieve high classification accuracy on leaf images, establishing the foundational methodology that subsequent studies have refined and deployed. This body of work has since grown substantially, driven by the urgent need to reduce crop losses in global food production systems.

A comprehensive review by Dolatabadian (2025), published in *Plant Pathology*, surveyed image-based disease detection methods across multiple crops and concluded that deep CNN models consistently outperform classical machine learning approaches in both accuracy and scalability. The review further emphasized that models must be tailored to their target region and crop species, noting that differences in environmental conditions can affect observable disease symptoms and incidence — a key consideration for a system like CORNGUARD targeting Bukidnon.

For corn specifically, recent comparative studies have confirmed the superiority of deep learning approaches. A 2025 study published in the journal *Artificial Intelligence in Applied*

Sciences trained and compared five prominent CNN architectures — DenseNet121, InceptionV3, MobileNetV2, ResNet50, and VGG16 — on the standard corn leaf disease dataset. Results showed that MobileNetV2 achieved the highest cumulative classification accuracy of 95% across all metrics and demonstrated the greatest computational efficiency, making it ideal for deployment on mobile devices. These findings are consistent with the model selection rationale underpinning CORNGUARD.

Pacal and Isik (2024), publishing in *Neural Computing and Applications*, further validated the use of CNN and Vision Transformer (ViT) models for precise corn leaf disease identification, confirming the robustness of CNN-based pipelines even under complex background and dilation challenges, achieving a test accuracy of 95.4%. A hybrid approach combining MobileNetV2 with a Vision Transformer, evaluated by Bekele et al. (2025) in the *New Zealand Journal of Crop and Horticultural Science*, achieved 96.73% accuracy on a dataset of over 4,000 images using the same four-class corn disease taxonomy relevant to CORNGUARD — Common Rust, Grey Leaf Spot, Healthy, and Northern Leaf Blight.

Kumar et al. (2025), in *Machine Learning with Applications*, reported that among multiple deep learning models evaluated for in-field maize leaf disease diagnosis, MobileNetV2 achieved the highest classification accuracy of 97%. Crucially, their best-performing models were integrated into a mobile application deployed on edge devices for real-time diagnosis by smallholder farmers in Uganda, demonstrating end-to-end field deployability analogous to the system architecture proposed for CORNGUARD.

Ashurov et al. (2025), publishing in *Frontiers in Plant Science*, proposed architectural enhancements to MobileNetV2 by integrating Squeeze-and-Excitation (SE) blocks and modified residual skip connections for plant disease detection. Evaluated on the PlantVillage dataset — the same dataset used as the training base for CORNGUARD — the enhanced model achieved 98% accuracy and an F1 score of 98.2% across 38 disease classes, including Common Rust, Gray Leaf Spot, and Northern Leaf Blight. The study confirms that MobileNetV2's depthwise separable convolution architecture serves as a strong foundation that can be further enhanced

without sacrificing computational efficiency, reinforcing CORNGUARD's model selection rationale.

Taken together, the literature establishes a strong empirical consensus: CNNs remain the backbone of reliable, deployable plant disease detection, and MobileNetV2 consistently ranks among the top-performing lightweight architectures for this task across multiple independent benchmark studies.

Target Diseases and Their Causal Agents

The three corn leaf diseases targeted by CORNGUARD are among the most economically significant fungal diseases affecting *Zea mays* globally and are particularly prevalent in tropical and subtropical agricultural regions such as Northern Mindanao.

Common Rust is caused by the obligate biotrophic fungus *Puccinia sorghi* Schwein., one of the most widespread pathogens affecting corn worldwide. The fungus reproduces through wind-dispersed urediniospores that germinate on leaf surfaces under conditions of high relative humidity and cool to moderate temperatures ranging from 16°C to 23°C (Patil et al., 2022). Upon germination, the spores penetrate the leaf epidermis through stomata, establishing infection within the mesophyll tissue. The disease manifests as small, circular to elongated reddish-brown pustules — known as uredia — scattered across both leaf surfaces. Under favorable environmental conditions, a single infection cycle can be completed in as few as seven days, allowing the pathogen to spread rapidly across a field before visible symptoms are widely recognized (Patil et al., 2022; Zhu et al., 2024). Yield losses attributed to Common Rust can reach up to 40% in highly susceptible varieties when infection occurs during early vegetative stages.

Gray Leaf Spot is caused by the fungus *Cercospora zeae-maydis* Tehon & E.Y. Daniels, a necrotrophic pathogen that thrives under warm, humid conditions with prolonged periods of leaf wetness — conditions that are characteristic of the tropical climate of Bukidnon and the broader

Northern Mindanao region (Ward et al., 1999; Dolatabadian, 2025). The pathogen overwinters in infected crop residue left on the soil surface and releases conidia that are dispersed by wind and rain splash to infect new leaf tissue. Infection typically initiates in the lower canopy and progresses upward as the season advances. Lesions appear as rectangular, tan-to-gray spots bounded by leaf veins, reflecting the pathogen's restricted growth pattern within vascular tissue. Under severe infection pressure, lesions coalesce to cover large portions of the leaf surface, leading to premature leaf senescence and significant reductions in photosynthetic capacity. Yield losses from Gray Leaf Spot can range from 20% to 50% depending on the timing and severity of infection (Ward et al., 1999; Khan et al., 2023).

Northern Leaf Blight is caused by *Exserohilum turcicum* (Pass.) K.J. Leonard & E.G. Suggs (formerly *Helminthosporium turcicum*), an ascomycete fungus that produces characteristic elongated, elliptical lesions often described as "cigar-shaped," measuring 2.5 to 15 centimeters in length (De Rossi et al., 2022). The pathogen spreads primarily through airborne conidia produced on the surface of existing lesions, with infection favored by moderate temperatures of 18°C to 27°C and high relative humidity or extended dew periods. Like *Cercospora zea-maydis*, *E. turcicum* can persist in infected plant debris between growing seasons, serving as a primary inoculum source for subsequent crops. Infection that establishes before or during the silking stage (R1) is particularly damaging, as it directly reduces grain fill and kernel development. Documented yield losses range from 15% to 50% depending on the growth stage at which infection occurs and the susceptibility of the host variety (De Rossi et al., 2022; Zhu et al., 2024).

All three pathogens are fungal in nature and share key epidemiological characteristics relevant to the CORNGUARD system: they produce visually distinct lesion patterns on corn leaf surfaces that are detectable through image-based analysis, they are capable of rapid spread under the humid tropical conditions prevalent in Bukidnon, and they represent the primary disease threats to smallholder corn farmers in the target region. These shared characteristics make them well-suited as classification targets for a CNN-based leaf image diagnostic system.

MobileNetV2: Lightweight Architecture for Mobile Deployment

MobileNetV2, introduced by Sandler et al. (2018) at Google, represents a landmark achievement in the design of computationally efficient neural networks. Its core innovations — depthwise separable convolutions and inverted residual blocks with linear bottlenecks — significantly reduce the parameter count and floating-point operations compared to standard CNNs, without proportionally sacrificing accuracy. These properties make it especially well-suited for deployment on resource-constrained mobile and embedded devices.

Alkanan and Gulzar (2024), publishing in *Frontiers in Applied Mathematics and Statistics*, demonstrated that an enhanced version of MobileNetV2 incorporating additional layers — Average Pooling, Flatten, Dense, and Dropout — achieved strong discriminative classification of corn diseases on a comprehensive dataset of 21,662 images. Their results confirmed the model's effectiveness when augmented through feature-level modifications and transfer learning from ImageNet, supporting the technical approach adopted by CORNGUARD.

The architectural extensibility of MobileNetV2 is further demonstrated by Ashurov et al. (2025), who enhanced the model with Squeeze-and-Excitation blocks and depthwise-modified residual skip connections, formalized mathematically as:

$$F_{out} = DwiseConv(x) + F_{SE}$$

where $DwiseConv(x)$ denotes a depthwise separable convolution applied to the input feature tensor and F_{SE} is the recalibrated feature tensor after the SE module. This modification achieved a 3% accuracy gain over the baseline MobileNetV2 without the SE block, and a 7% recall improvement when skip connections were present versus absent — confirmed through an ablation study. Their model size of 8.5 MB with a 12 ms inference time further validates that MobileNetV2-based architectures are viable for real-time, resource-constrained deployment, directly supporting CORNGUARD's target hardware profile.

A 2025 IEEE study on lightweight mobile deep learning for plant disease detection reported that MobileNetV2, after quantization and TensorFlow Lite conversion, could be

integrated into a Kotlin-based Android application with high AUC-ROC values of 0.99. The study demonstrated that model optimization steps including quantization and TFLite conversion effectively translated the model's in-lab accuracy into practical on-device performance, confirming the viability of the deployment pipeline planned for CORNGUARD.

In a systematic comparison of lightweight architectures, a 2025 study optimizing edge AI systems on PlantVillage data found that MobileNetV3-Small, an evolution of the MobileNetV2 lineage, could achieve 98.90% validation accuracy with only 1.25 million parameters and a model size under 5 MB after quantization — yielding inference times of 4.92 ms on standard CPU hardware. While CORNGUARD uses MobileNetV2, this trajectory of the MobileNet family confirms the upgrade path noted in the study's general objective and validates the architectural choice as future-proof.

Few-shot learning comparisons involving corn diseases published in PMC (2025) tested MobileNetV2 against ResNet-50 and DenseNet121, with MobileNetV2 achieving 94.12% accuracy — outperforming ResNet-50 at 92.34%. These consistent results across independent studies reinforce the conclusion that MobileNetV2 strikes the optimal balance between accuracy, model size, and inference speed for the target deployment environment of CORNGUARD.

Mobile Applications for Agricultural Disease Management

The development of mobile applications specifically targeting agricultural disease management has accelerated significantly in recent years, driven by the proliferation of smartphones in rural areas and the growing maturity of mobile deep learning frameworks. These applications represent a paradigm shift from cloud-dependent expert systems toward accessible, field-deployable tools for individual farmers.

Askale et al. (2025), publishing in *Plant Methods*, developed a novel real-time, user-friendly maize leaf disease detection and classification mobile application, noting that most prior research had focused exclusively on developing offline models without addressing accessibility or immediate feedback requirements. Their work compared VGG16, AlexNet, and

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ResNet50 architectures within a mobile deployment context, establishing that the combination of CNN classification and mobile interface design is both technically feasible and practically meaningful for field use.

Locally, the Bukidnon State University Corn Disease Detector — developed by Lapates (2024) and subsequently adopted by the Malaybalay City Agriculture Office — provides a critical regional benchmark. This system employs YOLOv8 for corn crop disease detection and has been validated in Mindanao agricultural conditions. CORNGUARD differentiates itself from this system through its focus on the MobileNetV2 architecture for lightweight, low-end device optimization and its per-farm disease history tracking capability — features that extend the diagnostic utility of the system for individual farmers.

Khan et al. (2023), in *Frontiers in Plant Science*, developed a mobile-based system for maize plant leaf disease detection using deep learning and demonstrated the practical applicability of AI-powered diagnostic tools for field-level disease identification. Their work emphasized the importance of addressing regional agricultural conditions and hardware limitations — both factors directly addressed in CORNGUARD's system design.

Smart farming trends in the Philippines documented by Farmonaut (2025) confirm that mobile and web applications are the primary interface through which Filipino farmers engage with digital agricultural advisory services. The same report identified disease detection as one of the most critical unmet needs for smallholder farmers managing less than two hectares, directly supporting the significance of the CORNGUARD system.

Community-Based Platforms and Peer-to-Peer Knowledge Sharing for Agricultural Disease Management

The integration of community-based communication features within agricultural technology platforms has emerged as a significant and evidence-supported strategy for improving disease management outcomes among smallholder farmers. This body of literature

directly underpins CORNGUARD's proposed community forum module, which allows farmers to report disease occurrences, warn neighboring farms of active outbreaks, and collaboratively discuss treatment approaches within a localized, app-integrated social platform.

One of the most well-documented precedents for this approach is Plantix, developed by Progressive Environmental & Agricultural Technologies (PEAT). The app's community forum feature — where farmers help one another diagnose plant diseases when the AI does not produce a conclusive result — generates approximately 15,000 community messages per day, with every uploaded photo geo-time-tagged to enable real-time disease outbreak tracking across regions (CGIAR, 2020). Plantix leveraged this crowdsourced spatial data to send push notifications alerting farmers in a given area when nearby users reported the same pest or disease — demonstrating a direct, practical model for community-driven early warning that CORNGUARD adapts for the Bukidnon corn farming context.

Rao et al. (2025), publishing in the Proceedings of the ERCICA 2024 Conference via Springer, proposed Crop Connect — a social media platform tailored to empower farmers worldwide through seamless communication, knowledge exchange, and collaboration. The system incorporates real-time crop monitoring, plant disease detection, and digital cooperative features — including coordinating group discussions and solving disease outbreaks — that the authors found significantly enhance efficiency and economic resilience among farming communities. Their work confirms the technical and practical feasibility of embedding social communication tools within AI-powered agricultural applications, which is the exact design principle applied in CORNGUARD's community module.

The scientific basis for community-driven agricultural knowledge sharing is further supported by An and Mahasuweerachai (2025), who published findings in the Journal of Agricultural Education and Extension (Taylor & Francis). Their study, drawing on an extensive social network analysis, found that interactions and knowledge sharing among farmers significantly influence the adoption of agricultural innovation — particularly for smallholder communities in developing regions. This finding directly supports CORNGUARD's community

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module rationale: by facilitating peer-to-peer disease discussions within the app, the platform increases the likelihood that AI diagnostic results and treatment recommendations will be adopted and acted upon by the wider farming community.

In the Philippine context specifically, Aquino and Concepcion (2025), publishing in the SSRG International Journal of Communication and Media Science, investigated the use of social media platforms by rubber farmers in Cotabato Province for obtaining information about Pestalotiopsis Leaf Fall Disease. Their study found that Facebook was the predominant platform used, with farmers primarily motivated by the need for urgent information gathering during active disease outbreaks. While the study confirmed that social media facilitates rapid access to disease information and visual aids, it also identified challenges including limited internet access, data costs, digital literacy gaps, and information credibility issues — all of which are directly addressed in CORNGUARD's community platform design and limitations.

More broadly, digital communities through social media, forums, and messaging applications have been found to enable farmers to collectively exchange practical advice and solve problems, creating translocal communities of practice that support experimentation and adaptation in agriculture (Ramavhale et al., 2024, as cited in ResearchGate, 2025). The FarmstandApp research synthesis (2026) further documents that when facing pest outbreaks, farmers increasingly crowdsource solutions by posting photos and descriptions in specialized digital groups — with collective problem-solving communities often identifying issues faster than individual consultation. CORNGUARD's forum is conceived as a localized equivalent of this model, specific to corn diseases and Bukidnon conditions.

AI Accessibility for Smallholder Farmers in Developing Regions

A recurring theme across the recent literature is the gap between the technical capabilities of AI-based disease detection systems and their practical accessibility to smallholder farmers in developing countries. Many state-of-the-art models achieve impressive accuracy on benchmark

datasets but have not been deployed in a form accessible to low-resource, low-connectivity agricultural environments.

Dolatabadian (2025) articulated this gap clearly, noting that despite remarkable advances in image-based disease detection, further investment in research and development and innovative approaches tailored to the needs and constraints of smallholder farmers in developing countries is essential. This includes consideration of limited computational resources, low-bandwidth environments, and the digital literacy challenges facing rural farming communities.

A 2025 AI and IoT-powered edge device study published in Nature Scientific Reports highlighted the mismatch between available AI solutions and smallholder farmer needs, observing that most existing IoT solutions depend on cloud computing, which is subject to latency, bandwidth constraints, and connectivity limitations for smallholder farmers with limited access to reliable internet infrastructure. This observation directly motivates CORNGUARD's offline-first, edge-inference architecture.

InsightNet research published in Plant Direct (2025) reaffirmed that a low-cost real-time inspection system with modest computational requirements would be highly beneficial for farmers, enabling them to quickly treat or isolate affected plants. The study determined that MobileNet-family models are both efficient and resource-light, supporting their use in real-time deployment on mobile or edge devices — a conclusion that validates CORNGUARD's core design assumptions.

Farmonaut's 2025 assessment of Philippine agricultural challenges identified land fragmentation and smallholder dominance as key structural constraints, with the majority of Filipino farmers managing less than two hectares and lacking access to advanced technologies. The report further identified pests and diseases as one of the most critical sources of crop losses, exacerbated by challenges of timely detection and effective management. CORNGUARD directly addresses this context by providing a portable, offline-capable tool designed specifically for the operational realities of Bukidnon corn farmers.

Collectively, the literature affirms that the intersection of lightweight deep learning, offline mobile deployment, and localized user design represents the most promising frontier in making AI-powered agricultural diagnostics genuinely accessible to smallholder farming communities. CORNGUARD is positioned at precisely this intersection, grounded in the empirical and technical foundations reviewed in this chapter.

TensorFlow Lite and On-Device Offline Inference

The deployment of deep learning models on resource-constrained mobile and edge devices has been made accessible through frameworks such as TensorFlow Lite (TFLite), which enables model quantization, size reduction, and optimized inference on Android and iOS devices. The ability to run inference entirely offline — without reliance on cloud connectivity — is a critical requirement for agricultural applications serving rural and remote farming communities where internet access is unreliable.

Goklani (2024), in a study published via SSRN, presented a mobile application for real-time plant disease detection using TensorFlow Lite and Flutter, demonstrating that CNNs converted to TFLite format can deliver accurate, responsive inference on standard smartphones. The study validated the feasibility of the offline-first architecture central to CORNGUARD's design.

A 2025 study presenting CropLeafNet, published in the International Journal of Information Technology, deployed a hybrid CNN architecture as a TFLite Android application for real-time, offline detection of multiple crop diseases, including cotton and castor leaf conditions. The system integrated the publicly available PlantVillage dataset with real-time field images, achieving robust performance and confirming the scalability of the TFLite-based pipeline for multi-crop, multi-environment scenarios. Their use of a sliding window mean average deviation preprocessing algorithm for brightness and color correction is particularly relevant to CORNGUARD's handling of variable field lighting.

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An optimized edge AI system evaluated in 2025, trained on 85,606 PlantVillage images spanning 37 disease classes and converted to TFLite with 8-bit quantization, achieved 99.65% training accuracy and 98.90% validation accuracy. The resulting model contained only 1.25 million parameters with a footprint under 5 MB — demonstrating that quantization does not materially degrade accuracy while dramatically reducing the computational demands on low-end devices such as those commonly used by Filipino smallholder farmers. CORNGUARD targets this same quantized TFLite deployment pipeline.

A lightweight mobile deep learning framework study published in IEEE Xplore (2025) confirmed that MobileNetV2, subjected to quantization and TFLite conversion and integrated into a Kotlin Android application, maintained strong classification performance with an AUC-ROC of 0.99. The study further validated that this pipeline is both practical and reproducible across different mobile hardware configurations, directly supporting the technical feasibility of CORNGUARD's offline inference module.

CHAPTER III

METHODOLOGY

This chapter presents the research design and development methodology adopted for the CORNGUARD project. The study follows a **Software Development Life Cycle (SDLC)** framework consisting of seven phases: Planning and Requirement Analysis, Defining Requirements, System Design, Coding and Implementation, Testing, Deployment, and Operations and Maintenance. This structured approach ensures that the development of the mobile application for real-time corn disease detection is systematic, traceable, and aligned with the project's objectives.

3.0 Development Model

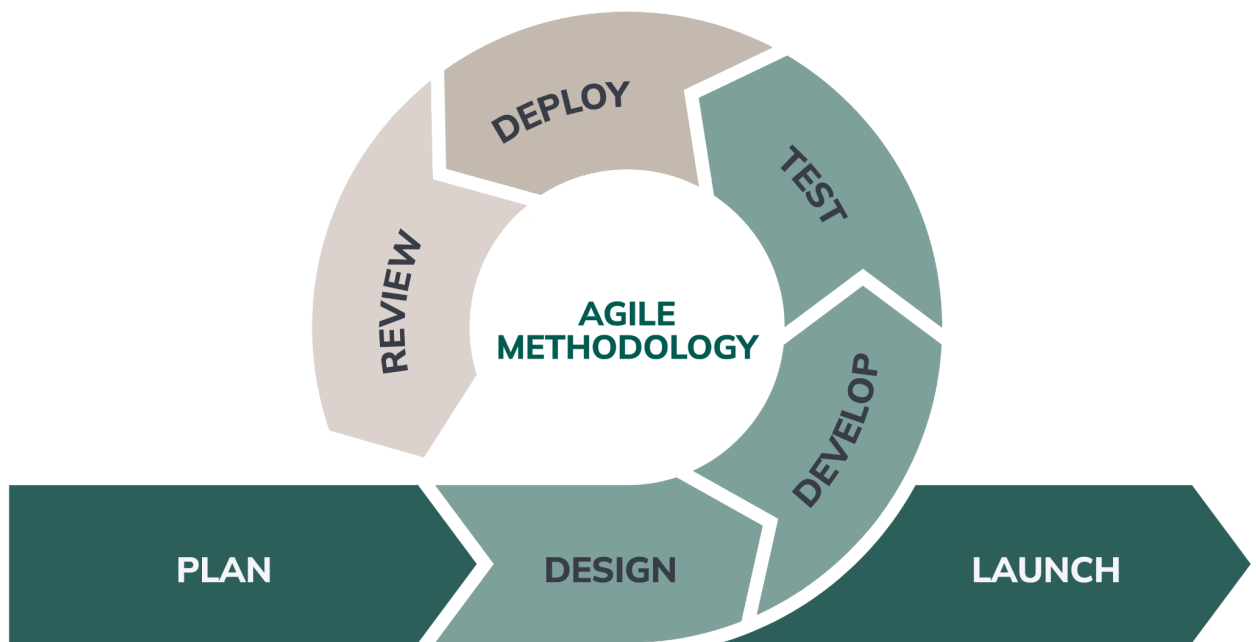


Figure 5: Agile Development Model

The development of CORNGUARD adopts the **Agile Software Development Life Cycle (SDLC)** as its overarching development philosophy. Agile is particularly well-suited for this

project because CORNGUARD is composed of multiple interdependent modules—the offline CNN inference engine, and the internet-connected community forum—each of which can be developed, tested, and refined in parallel sprints without blocking the others. The iterative nature of Agile also accommodates the experimental demands of deep learning development, where model accuracy must be evaluated repeatedly across training cycles and dataset adjustments before a satisfactory result is achieved. Furthermore, Agile's emphasis on continuous stakeholder feedback aligns directly with the study's farmer validation component, ensuring that the application is refined based on real field input from corn farmers in Bukidnon rather than finalized in isolation. By working in structured sprints, the development team can respond to findings from each phase—such as low model accuracy on a particular disease class or usability issues identified during User Acceptance Testing—without having to restart the entire development process, as would be required under a rigid linear model such as Waterfall.

3.1 Phase 1: Planning and Requirement Analysis

This is the discovery phase of CORNGUARD's development. It involves gathering requirements from primary stakeholders — specifically corn farmers in Bukidnon, local agricultural technicians, and extension workers — and determining the technical, operational, and economic feasibility of the proposed mobile-based disease detection system.

3.1.1 Requirements Gathering

The following data-gathering methods were employed to identify the system requirements:

Interviews: Structured interviews were conducted with local corn farmers in Bukidnon to understand current disease identification practices, challenges with manual inspection, and familiarity with mobile technology.

Observation: Field observations were performed on local corn farms to document real-world conditions such as lighting variability, leaf presentation, and the practical constraints of using a smartphone in the field.

Document Analysis: Existing agricultural reports from the Philippine Statistics Authority (PSA, 2023) and local government unit records were reviewed to quantify disease incidence and yield loss in the region.

Literature Review: Published studies on CNN-based plant disease detection (Mohanty et al., 2016; Kumar et al., 2025; Lapates, 2024) were analyzed to identify the most suitable deep learning architecture and dataset sources.

3.1.2 Dataset Collection and Image Acquisition

The dataset used for training and validating the CORNGUARD model consists of both publicly available and locally collected corn leaf images. Public datasets were sourced from repositories such as **PlantVillage** and **Kaggle**, while local data collection will be conducted in selected corn farming areas in Bukidnon.

A total of approximately **3,000–4,000** corn leaf images were utilized, covering four classes: Common Rust, Gray Leaf Spot, Northern Leaf Blight, and Healthy leaves. To improve model generalization and ensure real-world applicability, a portion of the dataset will be collected directly from multiple barangays within Bukidnon.

Local data collection covered at least 3–5 barangays, ensuring variation in environmental conditions and farming practices. Images were captured under different real-world conditions, including varying lighting (morning, afternoon, shaded), backgrounds (soil, sky, human hand), and leaf orientations.

Image acquisition was performed using commonly available Android smartphones with camera resolutions ranging from 8 MP to 48 MP. This approach

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ensures that the dataset reflects the actual image quality and conditions expected from end-users of the application.

To further enhance dataset diversity, images showing varying degrees of leaf damage and disease progression were included. Data augmentation techniques such as rotation, flipping, and brightness adjustment were also applied during preprocessing to simulate additional environmental variations.

DATASET NAME:	Corn or Maize Leaf Disease Dataset
SOURCE:	Kaggle
DOWNLOAD URL/ DOI:	https://www.kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset
TOTAL NUMBER OF IMAGES:	• Common Rust: 1,306 • Gray Leaf Spot: 574 • Northern Leaf Blight: 1,146 • Healthy: 1,162 Total: 4,188 images
VALIDATION & TESTING	The dataset will be divided into: • 70% Training Set — 2,931 images • 15% Validation Set — 628 images • 15% Testing Set — 629 images
NUMBER OF CLASSES:	4 classes (Common Rust, Gray Leaf Spot, Northern Leaf Blight, Healthy)
IMAGE FORMAT:	jpg/png

IMAGE RESOLUTION:	Original images vary in size (approximately 300×300 pixels). For model training, all images are resized to 224×224 pixels.
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3.1.3 Data Preprocessing

Before being fed into the MobileNetV2 model, all corn leaf images will undergo a series of preprocessing steps to ensure consistency, reduce noise, and optimize model performance. These steps are critical for achieving reliable disease classification in real-world field conditions where lighting, background, and image quality vary significantly.

Preprocessing Steps

Step 1: Image Loading. All images will be loaded in RGB color mode to ensure three color channels (red, green, blue) for processing. Images with file extensions .jpg and .jpeg were both handled, as the dataset contained mixed file naming conventions.

Step 2: Resizing. Each image will be resized to 224×224 pixels. This specific dimension is required because MobileNetV2 expects a fixed input size of 224×224 pixels. Resizing also reduces computational requirements while preserving sufficient detail for detecting disease symptoms such as pustules, lesions, and spots.

Step 3: Normalization. Pixel values will be normalized using MobileNetV2's built-in preprocessing function, which scales pixel values from the original range of [0, 255] to [-1, 1]. This normalization convention is specific to MobileNetV2's ImageNet pretraining and ensures that input values are consistent with what the model expects during both training and inference.

Step 4: Data Augmentation (Training Set Only). To increase dataset diversity and reduce overfitting, the following random augmentations will be applied only to the training set. The validation and test sets received only normalization — no augmentation.

- Random horizontal flip — to account for leaf orientation variations
- Random vertical flip — to account for varying image capture angles in the field
- Random rotation (± 15 degrees) — to account for different leaf angles
- Random zoom ($\pm 10\%$) — to account for varying camera distances
- Random brightness adjustment (80% to 120%) — to account for different lighting conditions (morning, afternoon, shaded)

Step 5: Class Imbalance Handling. Due to the underrepresentation of Gray Leaf Spot in the dataset (574 images — approximately 2.28 times fewer than Common Rust), balanced class weights will be computed using `sklearn`'s `compute_class_weight` function and applied during model training. This ensures that the model penalizes misclassification of the minority class more heavily, preventing bias toward the more frequent classes.

3.2 Feasibility Study

A three-part feasibility study was conducted to confirm the viability of the project prior to system design and development.

3.2.1 Technical Feasibility

The CORNGUARD system relies entirely on mature, well-documented, and freely available technologies. MobileNetV2, TensorFlow Lite, and the Android SDK are all widely supported and compatible with the team's existing technical skills in Python, Kotlin, and machine learning. The PlantVillage dataset and locally captured corn leaf images from Bukidnon provide a sufficient training base for the four target disease classes.

3.2.2 Operational Feasibility

A key finding from farmer interviews is that many target users own low-end Android smartphones but lack reliable internet access. CORNGUARD's offline-first architecture — where CNN inference is performed locally via TensorFlow Lite with no internet dependency — directly addresses this constraint. The UI is designed to be minimal and icon-driven to accommodate varying levels of digital literacy among rural farming communities.

3.2.3 Economic Feasibility

The entire software stack is open-source (₱0 in license costs). Model training is performed on Google Colab (free tier). The locally captured image dataset incurs only travel costs for field data collection. Given that Bukidnon accounted for 68.1% of Region X's total corn production of 1,520,889 metric tons in 2023 and topped national agricultural production value at ₱155.28 billion (PSA, 2023), the economic stakes of disease-related crop losses are substantial. Even modest improvements in detection accuracy and treatment timing yield a considerable positive return on investment.

3.3 Defining Requirements (Software Requirements Specification)

The information gathered during Phase 1 was formalized into a Software Requirements Specification (SRS) document. The SRS serves as the authoritative reference for all subsequent design and development decisions, ensuring that CORNGUARD is built to meet the verified needs of its target users.

3.3.1 Functional Requirements

Functional requirements define what CORNGUARD must do. The core functional requirements derived from stakeholder interviews and literature analysis are as follows:

- **Disease Classification:** The system shall capture a corn leaf image and return a classification result identifying the disease as Common Rust, Gray Leaf Spot, Northern Leaf Blight, or Healthy, along with a confidence score.
- **Offline Inference:** The disease detection module shall operate without an active internet connection, using an on-device TensorFlow Lite model.
- **Treatment Recommendations:** The system shall provide general agronomic treatment recommendations based on the classification result.
- **Community Forum:** The system shall allow registered users to post disease reports with optional photos, view reports from nearby farmers, and participate in discussion threads organized by disease type and location (requires internet connectivity).
- **Disease History:** The system shall maintain a per-farm log of past diagnoses to support trend monitoring by the user and local agricultural offices.
- **GIS-Based Disease Mapping:** The system shall display disease occurrences on a map using location-based data to help users identify nearby cases and monitor potential outbreak areas.

3.3.2 Non-Functional Requirements

Non-functional requirements define how CORNGUARD performs. The primary non-functional requirements are:

- **Classification response time:** result must appear within 3 seconds of image capture on a mid-range Android device.
- **APK size:** the total application package must not exceed 50 MB to remain installable on low-storage devices.
- **Model accuracy:** the MobileNetV2 TFLite model must achieve a minimum of 90% accuracy on the held-out test set.
- **Usability:** the interface must be navigable by a user with minimal digital literacy, requiring no more than three taps to reach the camera capture screen from the home screen.

3.4 Phase 2: System Design

This phase translates the SRS into a concrete architectural blueprint for CORNGUARD. The design work focuses on the overall system structure and defines the architecture, components, and their interactions.

3.4.1 System Architecture

The CORNGUARD system architecture illustrates the interaction between users, the mobile application, network services, and cloud infrastructure. The system has two users: the Farmer and the Admin.

Farmers use the CORNGUARD Android application, which contains two main modules: the Offline AI Detection module and the Community Module. The Offline AI Detection module uses

a TensorFlow Lite model for image capture, disease detection, and treatment recommendations without requiring internet access. The Community Module provides online features such as forum posts, disease reports, discussions, and notifications.

When internet connectivity is available, the application connects to the Firebase Server through the Internet/API layer. Firebase handles authentication, data synchronization, cloud storage, and push notifications. The Cloud Database stores user data, diagnosis records, community posts, comments, and notifications, while Cloud Storage manages uploaded images and attachments.

The Admin accesses the Firebase backend to manage users, moderate community content, and monitor diagnosis reports and notifications.

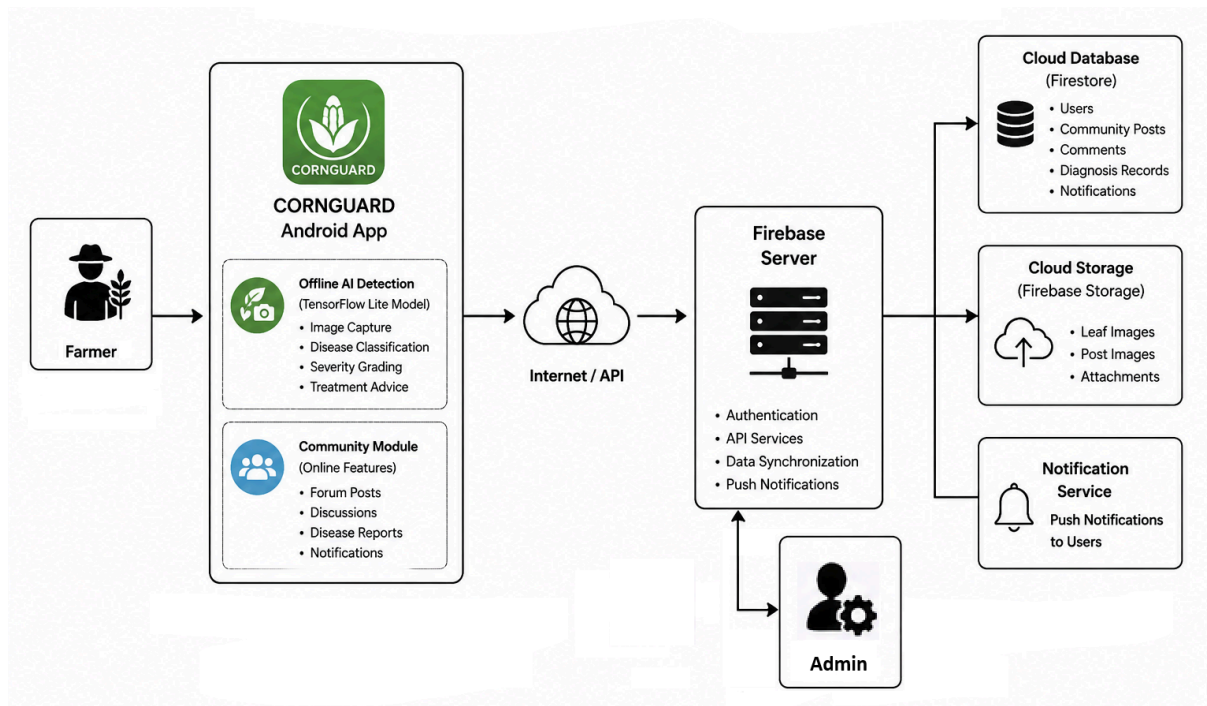


Figure 6: System Architecture of CORNGUARD.

3.4.2 Data Flow Diagrams (Level 0 and Level 1)

The DFD illustrates how image data flows from the farmer's camera input, through the TFLite inference engine, to the classification output and optional community post upload.

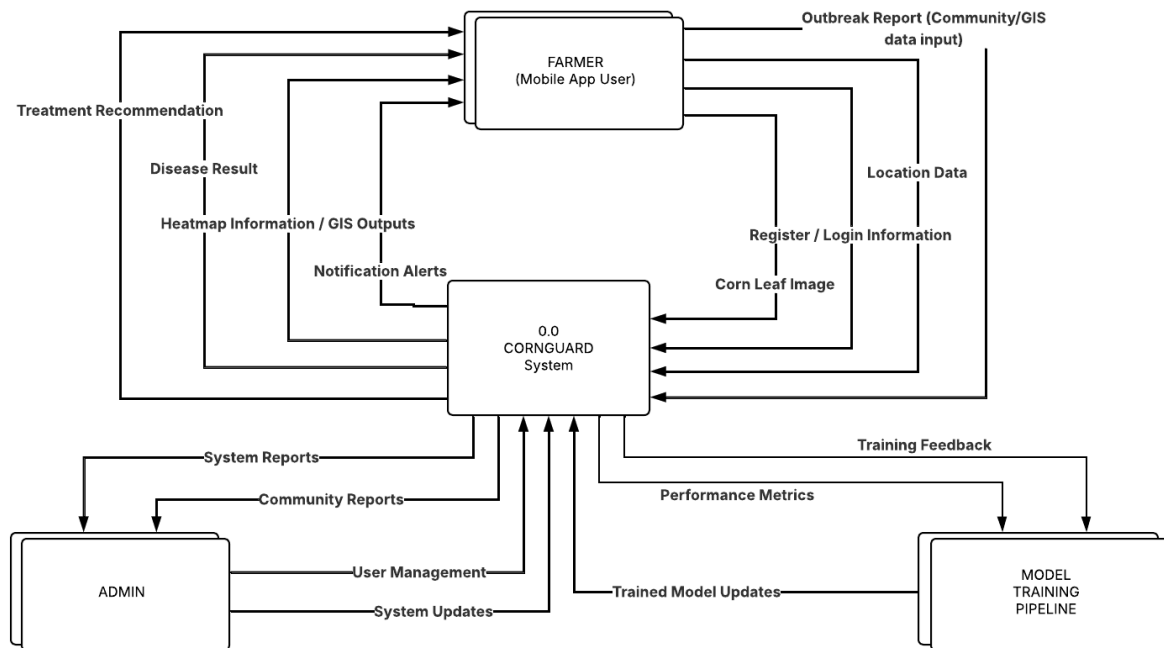


Figure 7: Level 0 Data flow Diagram(Context)

The Level 0 Data Flow Diagram presents the overall flow of data between CORNGUARD and its external entities. Farmers provide leaf images and user data to the system and receive diagnosis results, treatment recommendations, and disease history information. Administrators manage system updates and user-related reports. The model training pipeline supplies trained machine learning updates and receives performance metrics for continuous model improvement.

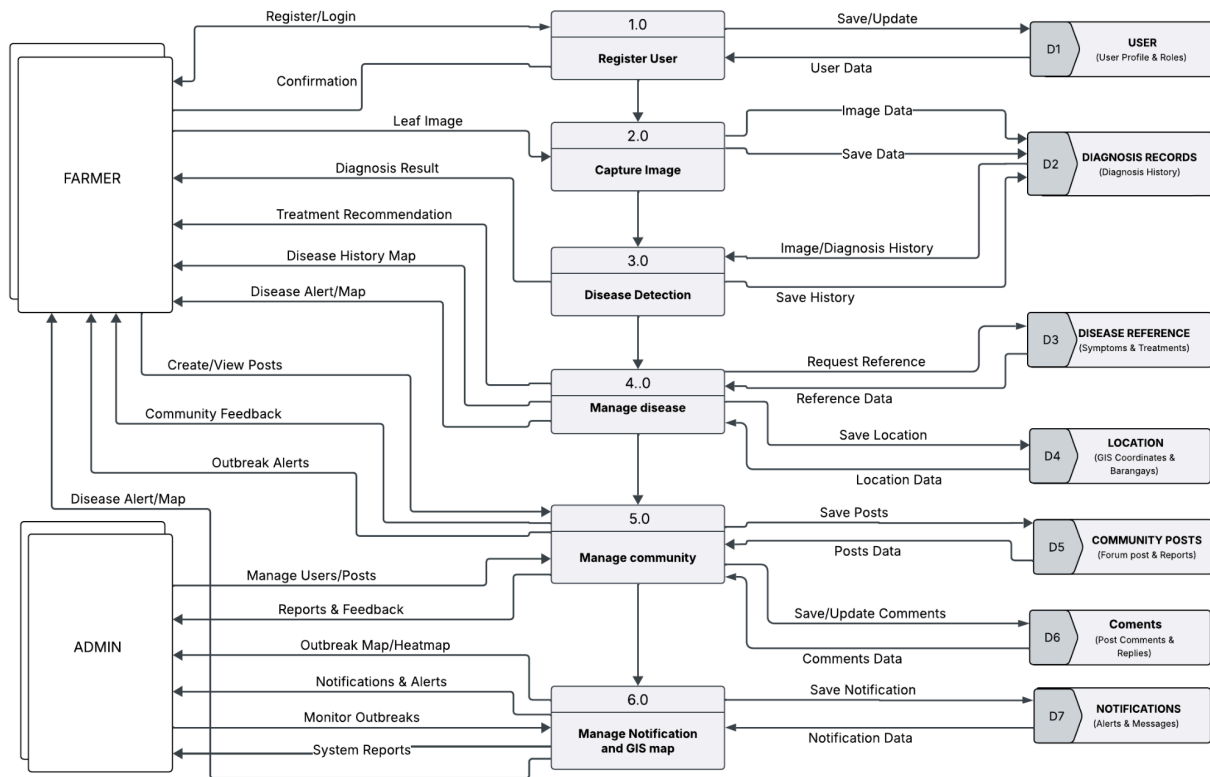


Figure 8: Level 1 Data flow Diagram

The Level 1 Data Flow Diagram illustrates the internal processes of CORNGUARD and their interactions with system data stores. The system is divided into major processes including Image Capture and Preprocessing, Disease Detection, Disease Management, Community Management, and GIS Mapping and Notification Management. These processes interact with the USER, DIAGNOSIS_RECORDS, DISEASE_REFERENCE, LOCATION, COMMUNITY_POSTS, and NOTIFICATIONS data stores to manage disease detection, treatment recommendations, community discussions, geographic disease mapping, and notification services within the application.

3.4.3 Use Case Diagram

The use case diagram identifies the primary actors — Farmer and System Administrator — and their interactions with CORNGUARD's modules.

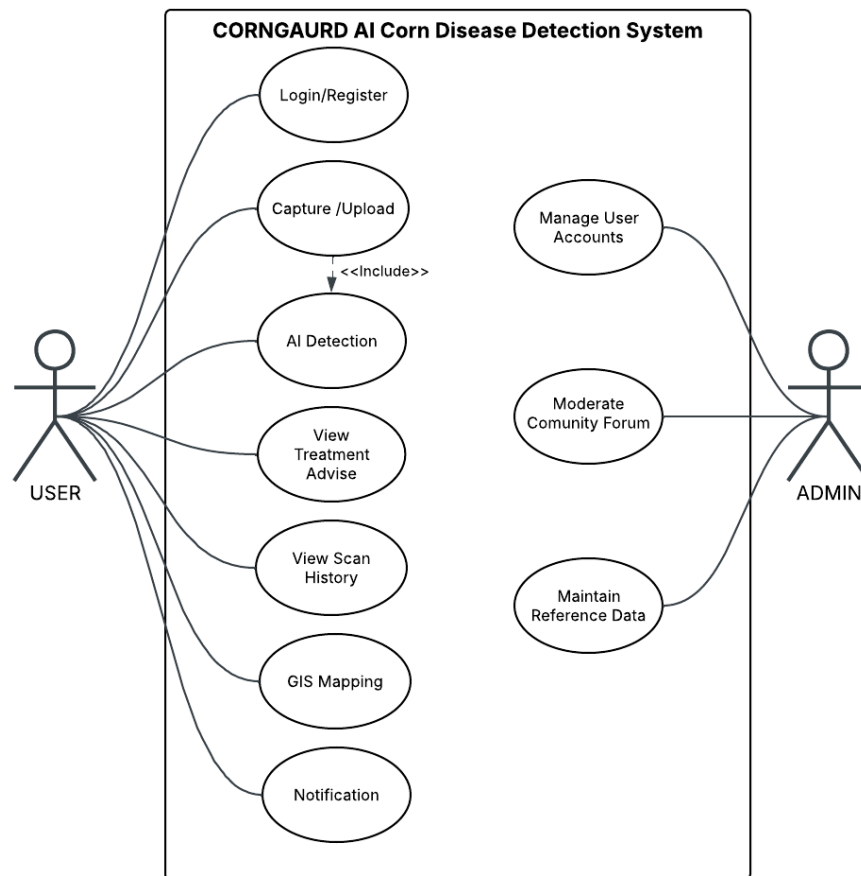


Figure 9: Use Case Diagram of CORNGUARD

Farmer Interactions: The Farmer serves as the primary data source, initiating the Capture/Upload Leaf Image process. This triggers the AI Disease Detection (using the CNN model) and provides the user with Treatment Advice and access to the GIS Disease Mapping.

System Administrator Interactions: The Administrator operates at the backend level, focusing on Managing User Accounts, Moderating the Community Forum, and Maintaining System Reference Data (such as updating the disease knowledge base).

3.4.4 Database Schema

The application database stores the following primary entities: User (farmer profile), DiagnosisRecord (per-farm disease log), CommunityPost (forum entries), and DiseaseReference (agronomic treatment guidelines). Relationships between entities are defined through foreign keys, and the schema is implemented using SQLite for the offline local database and a relational cloud database for the community forum backend.

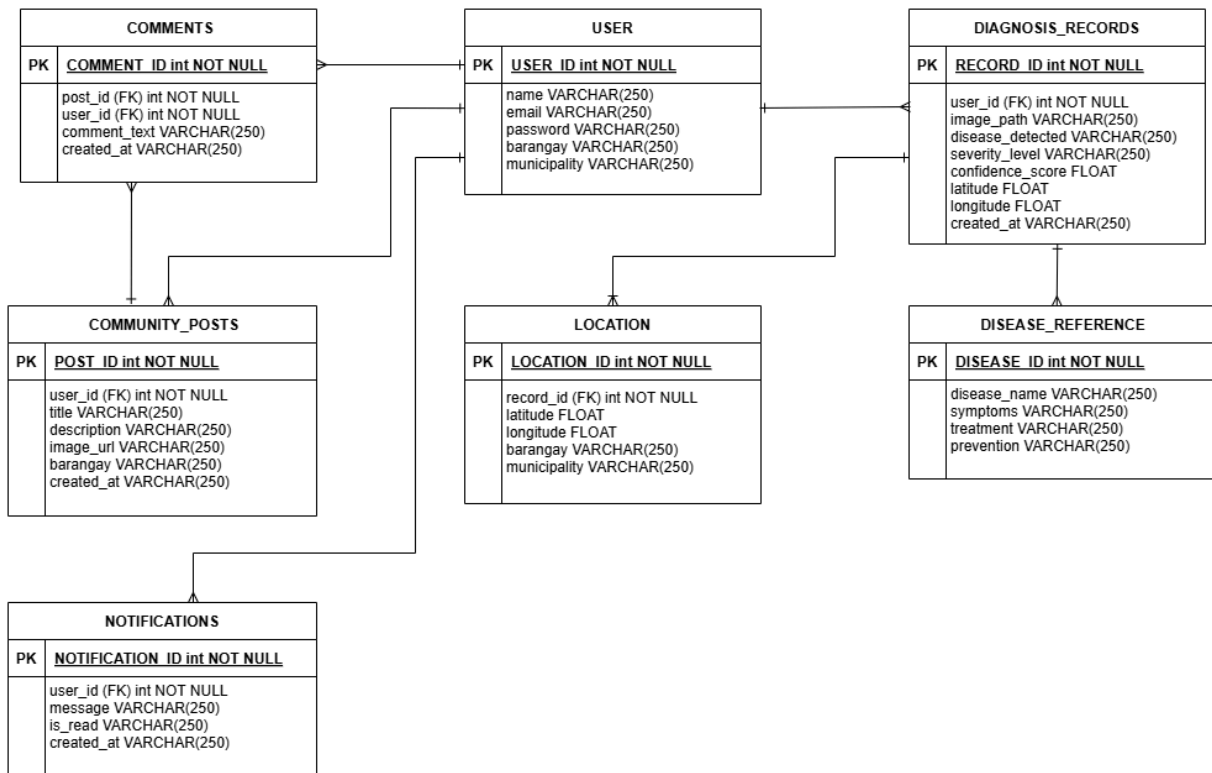


Figure 10: Entity Relationship Diagram of CORNGUARD

The CornGuard ERD represents a robust data architecture designed to support AI-driven disease detection and geographic monitoring. At its center, the USER table manages profile and regional information, serving as the primary link to all other system activities, such as posting in the community or receiving alerts. The core functionality is captured in the DIAGNOSIS_RECORDS and DISEASE_REFERENCE tables, which log AI scan results—including confidence scores and image paths—while providing users with essential symptoms and treatment advice. To meet the GIS requirements, a specialized LOCATION table connects to these records to store precise spatial data, specifically latitude and longitude coordinates and barangay-level details, which allows the system to generate real-time outbreak maps. Finally, the interaction layer—composed of COMMUNITY_POSTS, COMMENTS, and NOTIFICATIONS—facilitates social engagement and expert verification, ensuring that farmers and technicians can communicate effectively regarding crop health.

3.4.5 Algorithm Used

MobileNetV2

The CORNGUARD system utilizes the **MobileNetV2** architecture as its core deep learning algorithm for corn leaf disease classification. MobileNetV2 is a lightweight Convolutional Neural Network (CNN) specifically designed for mobile and embedded devices, making it highly suitable for real-time, on-device inference in low-resource environments such as rural farming areas.

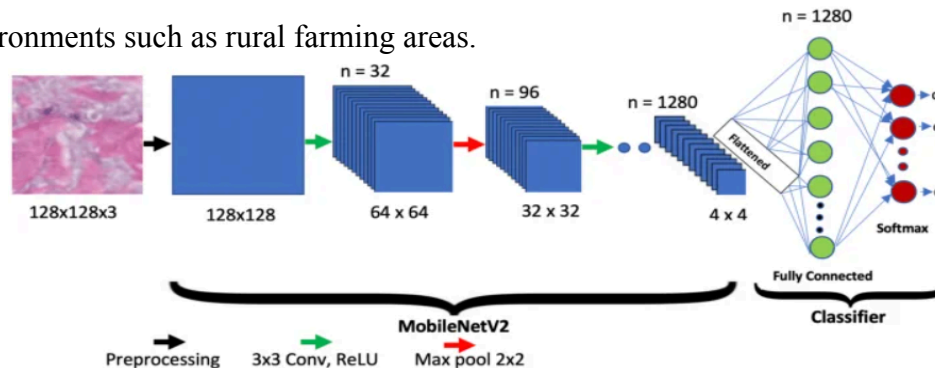


Figure 11. MobileNetV2 architecture

The model is implemented using transfer learning, where a pre-trained MobileNetV2 model trained on the ImageNet dataset is fine-tuned using a labeled dataset of corn leaf images. This approach significantly reduces training time while improving classification accuracy.

Further, Ashurov et al. (2025) demonstrated that the efficiency of MobileNetV2 can be enhanced through grouped depthwise convolutions with a group count of G , which reduces the number of parameters while preserving inter-channel information flow — an architectural direction that informs the potential optimization path for future iterations of CORNGUARD.

The processing pipeline of the MobileNetV2 model in CORNGUARD consists of the following stages:

- Image Acquisition:

The user captures a corn leaf image using the mobile device camera or selects an existing image from the gallery.

- Preprocessing:

The input image is resized to 224×224 pixels to match the required input dimensions of MobileNetV2. Pixel values are normalized to ensure consistent input scaling.

- Feature Extraction:

The MobileNetV2 architecture extracts relevant visual features such as color variation, lesion patterns, and texture differences associated with specific corn diseases.

- Model Inference:

The processed image is passed through the trained model, which outputs classification probabilities for each category: Common Rust, Gray Leaf Spot, Northern Leaf Blight, and Healthy.

- **Output Generation:**

The system displays the predicted disease class along with a confidence score and triggers the treatment recommendation module.

The use of MobileNetV2 ensures efficient computation, reduced memory usage, and fast inference time, enabling the application to function effectively on low-end Android devices without requiring internet connectivity.

The preprocessing pipeline adopted by CORNGUARD — resizing to 224×224, normalization of pixel values to [0,1], and data augmentation through rotation, flipping, and brightness adjustment — is consistent with the methodology validated by Ashurov et al. (2025) in their MobileNetV2-based plant disease detection study. Their normalization formula, defined as $x_{normalized} = x / 255.0$, and augmentation operations including random rotation by angle θ and luminance scaling by factor α (where $0.8 \leq \alpha \leq 1.2$) are mathematically equivalent to the preprocessing steps implemented in CORNGUARD, providing additional empirical grounding for these design choices.

Convolutional Neural Network

Convolutional Neural Networks (CNNs) are a specialized class of deep learning algorithms derived from the Multi-Layer Perceptron (MLP), specifically designed to process structured grid data such as two-dimensional images (LeCun et al., 1998). Unlike traditional machine learning approaches that require manual feature extraction, CNNs automatically learn hierarchical visual features directly from raw pixel input, making them particularly effective for image classification tasks (Mohanty et al., 2016). The supervised learning method used by CNNs involves training the model on labeled datasets, where the network learns to map input images to their corresponding output classes through iterative weight optimization.

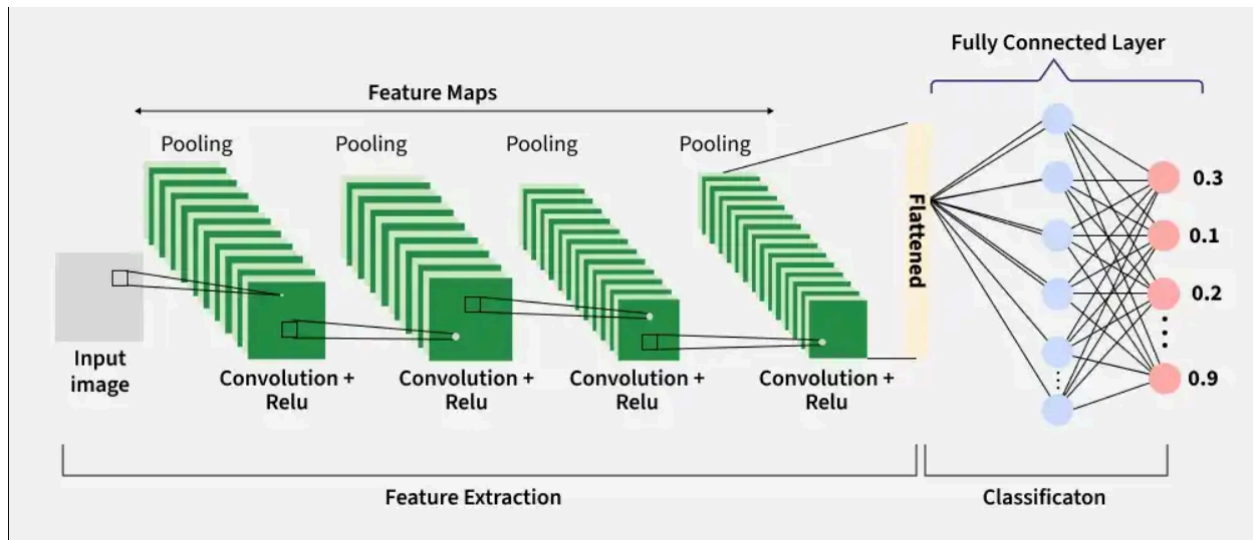


Figure 12. Convolutional Neural Network architecture

The architecture of a CNN consists of two primary components: the feature learning stage and the classification stage. In the feature learning stage, the input image passes through a series of convolutional layers, each followed by a Rectified Linear Unit (ReLU) activation function that introduces non-linearity into the model (Nair & Hinton, 2010). After each activation, a pooling layer reduces the spatial dimensions of the feature maps, retaining the most significant features while reducing computational cost and controlling overfitting (LeCun et al., 1998). This process is repeated across multiple convolutional blocks, allowing the network to progressively learn increasingly complex visual patterns — from simple edges and color gradients in early layers to disease-specific lesion shapes and textures in deeper layers (Ferentinos, 2018).

CNNs are particularly well-suited for corn disease detection because their spatial invariance allows the model to identify disease symptoms regardless of where they appear on the leaf surface (Too et al., 2019). Their robustness to real-world variations in lighting, camera angle, and leaf orientation makes them a practical choice for field-based mobile diagnosis applications (Barbedo, 2018). Mohanty et al. (2016) demonstrated that CNN-based models trained on the PlantVillage dataset — the same dataset used as the

training base for CORNGUARD — could achieve classification accuracy exceeding 99% under controlled conditions, establishing the foundational methodology for image-based plant disease detection. More recent studies have further confirmed that CNN architectures remain the most reliable and deployable approach for plant disease classification across diverse crop species and environmental conditions (Dolatabadian, 2025; Kumar et al., 2025).

In this study, the CNN is applied to corn leaf images as input and produces disease classification results as output. The trained model is deployed on a mobile device using TensorFlow Lite for real-time, offline field diagnosis by farmers in Bukidnon, making it a practical and accessible solution for on-site corn disease detection (Goklani, 2024).

The CNN detects corn disease through a seven-step process. The input layer first receives a corn leaf image of $224 \times 224 \times 3$ RGB pixels, which is then passed to the convolutional layer where filters are applied to detect low-level features such as leaf edges, texture patterns, and color variations including yellow spots and brown lesions. The ReLU activation function then introduces non-linearity, allowing only positive features to pass through. A pooling layer follows, reducing spatial dimensions and making detection robust to leaf rotation and position. Multiple convolution and pooling cycles are subsequently repeated to capture mid-level features such as lesion shapes and spot clusters, and ultimately high-level disease-specific patterns. The fully connected layer then flattens and consolidates all extracted features before passing them to the Softmax output layer, which produces a probability distribution across the disease classes — Common Rust, Gray Leaf Spot, Northern Leaf Blight, and Healthy.

3.4.6 Hardware Requirements

The CORNGUARD system is designed to operate on commonly available consumer devices, particularly low- to mid-range Android smartphones used by smallholder farmers.

Minimum Requirements:	
Processor:	Dual-core CPU (1.4 GHz or higher)
RAM:	4 GB
Storage:	At least 500 MB free space
Camera:	Built-in smartphone camera
Operating System:	Android 8.0 (Oreo) or higher

Recommended Requirements:	
Processor:	Octa-core CPU (2.0 GHz or higher)
RAM:	6 GB or higher

Storage:	1 GB or more free space
Camera:	8 MP or higher for improved image quality

The performance of the disease detection module, particularly inference speed, is dependent on the processing capability of the device. However, the use of TensorFlow Lite ensures optimized performance even on entry-level hardware.

3.4.7 Software Stack

The development of the CORNGUARD system utilizes a combination of software tools and frameworks across different layers of the application:

Frontend (Mobile Application):	
Kotlin (Android development)	The primary programming language for Android development, chosen for its concise syntax, null safety, and seamless integration with TensorFlow Lite.
XML (User Interface design)	Used to design the user interface layouts, providing a clear separation between the visual components and the application logic.

Backend (Community Module):

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Firebase Firestore / Realtime Database	A cloud NoSQL database that stores community forum posts, comments, and user profiles with real-time synchronization.
Firebase Authentication	Manages user registration and login, ensuring secure access to the community features.
Firebase Cloud Messaging (for notifications)	Delivers push notifications to farmers when disease outbreaks are reported in their area.

Machine Learning and AI:	
Python (model development)	Used for developing and training the MobileNetV2 deep learning model.
TensorFlow / Keras (model training)	Deep learning frameworks used to build, train, and fine-tune the CNN model for corn disease classification.
TensorFlow Lite (model deployment on mobile)	Converts the trained model into a lightweight format optimized for on-device inference on Android smartphones.
OpenCV (image preprocessing)	Handles image preprocessing tasks such as resizing, noise reduction, and color space conversion.
NumPy (numerical operations)	Performs numerical operations and array manipulations during model training and preprocessing.

Development Tools:	
Android Studio (mobile development environment)	The official integrated development environment (IDE) for building the CORNGUARD mobile application.
Google Colab (model training)	A cloud-based Jupyter notebook environment used to train the MobileNetV2 model using free GPU resources.
GitHub (version control)	A version control platform used for code collaboration, tracking changes, and managing project versions.

This software stack enables efficient model development, seamless mobile integration, and scalable cloud-based communication for the community platform.

3.4.8 Software Requirements

1. Functional Requirements (FR)

These requirements define the specific features and direct interactions users (farmers, agronomists, or researchers) will have with your mobile, web, or edge-based application.

Requirement ID	Module / System Feature	Requirement Description	Priority
FR-01	Image Capture / Upload	The system shall allow users to take a real-time photo of a corn leaf via an integrated device camera or upload an existing image from a local gallery.	High
FR-02	Image Preprocessing	The system must automatically resize, format, and normalize input images to match the exact input shape tensor (224 $\times$ 224 $\times$ 3) expected by the CNN model.	High

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FR-03	Disease Inference	The system shall feed the processed tensor into the trained CNN model to classify the leaf into one of four classes: Blight, Common Rust, Gray Leaf Spot, or Healthy.	High
FR-04	Inference Output Display	The system must display the final classification label along with a prediction confidence percentage score (e.g., Common Rust: 94.2%).	High
FR-05	Treatment Suggestion	Upon detecting a disease, the system shall fetch and display actionable mitigation steps, suggested fungicides, or cultural management practices from a local database.	Medium
FR-06	Historical Log / History	The system should save past scan results (including timestamps, locations, and detected labels) to a local or cloud-based user history log for tracking crop health over time.	Medium
FR-07	Offline Functionality	The system should be capable of running localized inferences using optimized architectures (like MobileNet or ONNX Runtime) even when internet connectivity is completely unavailable in rural fields.	High

2. Non-Functional Requirements (NFR)

These requirements specify the operational criteria, constraints, and quality metrics that your application must maintain.

Requirement ID	Category	Requirement Description	Priority
NFR-01	Performance	The CNN model inference pipeline must process an image and return classification results to the user UI in less than 2.0 seconds under normal operating conditions.	High
NFR-02	Accuracy Threshold	The embedded machine learning model must maintain a minimum target accuracy of 90.0% and an F1-Score of 0.88 across all testing validation subsets before production deployment.	High
NFR-03	Portability / Scalability	The mobile interface client application must run seamlessly on Android devices (API Level 24 and above) and/or cross-platform web browsers to ensure wide accessibility for smallholder farmers.	Medium
NFR-04	Usability	The user interface must feature localized text or simplified, highly visual, iconography so that farmers	High

		without technical or background computer literacy can operate the scan diagnostic tools easily.	
NFR-05	Reliability & Availability	The system database housing treatment resources should achieve 99.5% uptime availability during peak crop harvesting and inspection seasons.	Medium
NFR-06	Security / Privacy	Any user account information and geotagged field location data logged during crop scanning must be encrypted at rest and during transit using HTTPS/TLS protocols.	High

3.5 Phase 3: Testing

The testing phase will evaluate whether CORNGUARD meets both its technical performance requirements and its practical usability standards in real-world field conditions. Testing will be conducted across three levels: model evaluation, system testing, and User Acceptance Testing (UAT).

3.5.1 Model Evaluation

The MobileNetV2 classification model will be evaluated on a held-out test dataset that will not be used during training or validation. The test set will comprise a minimum of 20% of the total collected dataset, stratified across all four disease classes—common rust, gray leaf spot, northern leaf blight, and healthy—to ensure balanced evaluation.

Performance will be measured using four standard metrics: accuracy, precision, recall, and F1-score. The minimum acceptable accuracy threshold will be set at 90%, as defined in the non-functional requirements. A confusion matrix will also be generated to identify which disease classes are most frequently misclassified, guiding any necessary model refinements.

1. Accuracy

Accuracy measures the overall percentage of correctly classified images.

Formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Accuracy determines how well the model performs across all disease classes.

2. Precision

Precision measures how many predicted positive classifications are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

High precision indicates that the model produces fewer false disease detections.

3. Recall

Recall measures the model's ability to correctly identify diseased leaves.

$$Recall = \frac{TP}{TP + FN}$$

High recall is important to minimize missed disease cases.

4. F1-Score

The F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

This metric provides a balanced measurement of model performance.

5. Confusion Matrix

A confusion matrix will also be generated to visualize the classification performance of the model across the four classes:

- Common Rust
- Gray Leaf Spot
- Northern Leaf Blight
- Healthy

6. Validation and Testing Procedure

The validation dataset will be used during training to monitor overfitting and optimize model parameters, while the testing dataset will be used for final performance evaluation.

7. TensorFlow Lite Performance Evaluation

After conversion to TensorFlow Lite, the optimized model will be also evaluated on Android devices based on:

- Inference speed
- Response time

- APK size
- Offline functionality

This ensures that the model remains suitable for low-end Android smartphones used by farmers in Bukidnon.

3.5.2 System Evaluation and Testing

This section describes how the overall performance, functionality, and usability of the CORNGUARD system will be evaluated as a complete, integrated application. Unlike model evaluation, which focuses solely on the AI classifier's predictive accuracy, system evaluation examines how all components — the disease detection engine, treatment recommendation system, GIS mapping, and community forum — work together under real-world usage conditions.

A. System Testing

System testing will be conducted to verify that all individual modules and integrated components function correctly and reliably. Unit testing will be applied to each core module, including the disease detection engine, treatment recommendation system, and Firebase-based community forum, to confirm that each operates as intended in isolation. Integration testing will follow to ensure proper data flow between system components — for instance, verifying that a completed scan correctly passes the detected disease type and device location to both the local history log and the community post pre-population feature.

Offline functionality testing will confirm that core features — specifically disease detection, history logging, and treatment recommendations — remain

fully operational without any internet connection, in line with the system's primary design requirement for rural field use. Performance testing will assess whether the application delivers classification results within three (3) seconds on a mid-range Android device and maintains acceptable stability and application size under sustained use.

B. System Evaluation

System evaluation will assess the application's overall quality across three dimensions: performance, functionality, and usability.

In terms of system performance, the application will be evaluated for response time, operational stability, and processing efficiency. It is expected that classification results will be delivered within three (3) seconds under normal field conditions, without crashes or performance degradation across repeated use sessions.

For system functionality, evaluation will confirm that all features operate correctly and are properly integrated. This includes disease detection, treatment recommendation, offline storage, GIS-based outbreak mapping, and the community forum. Data flow between all modules will be verified to ensure that outputs from one component correctly inform downstream features.

C. Live Environment Testing with Farmers

To validate CORNGUARD under real-world agricultural conditions, system testing will be conducted in a live field environment with the direct participation of corn farmers from Bukidnon. The evaluation will involve a target of **30 farmer-respondents**, selected from corn-farming communities in the province. This sample size follows the approach commonly used in UAT studies for agricultural mobile applications in the Philippine context, where a minimum

of 20 to 30 end-user respondents is considered sufficient to identify critical usability issues and generate statistically meaningful satisfaction data (ISO 9241-11; Tullis & Albert, 2013).

During live testing, each farmer-respondent will be asked to perform a structured set of tasks using the installed CORNGUARD application on a provided or personally owned Android device. These tasks include capturing a corn leaf image, viewing the disease detection result, and reading the treatment recommendation, navigating to the disease history and GIS outbreak map, and creating a post in the community forum. Observations will be recorded regarding task completion rate, time on task, and any errors or points of confusion encountered during use.

D. IT Expert Evaluation

In addition to farmer testing, a separate evaluation panel composed of IT experts — specifically faculty members or practitioners with background in mobile systems development, software engineering, or agricultural technology — will assess the system using a structured technical evaluation instrument. The IT expert evaluation is necessary to validate aspects of the system that go beyond end-user experience, including code functionality, system architecture, integration completeness, and technical adherence to design specifications. A minimum of three (3) IT evaluators will be engaged for this purpose, consistent with the expert review approach adopted in similar capstone and applied research studies.

E. Evaluation Instruments and Survey Questions

Both farmer-respondents and IT evaluators will complete structured questionnaires following their respective testing sessions. The farmer usability questionnaire will be adapted from a System Usability Scale (SUS) framework

and will include items measured on a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree). Representative items include:

1. The application was easy to use even without prior training.
2. The disease detection results were clearly presented and easy to understand.
3. The treatment recommendation provided was helpful and relevant to what I observed.
4. I was able to complete the scan and get a result without needing the internet.
5. I would use this application again when I notice disease symptoms on my corn plants.
6. The community forum feature would be useful for sharing disease information with other farmers.
7. I am satisfied with the overall performance of the CORNGUARD application.

The IT expert questionnaire will cover functionality, reliability, usability, efficiency, and maintainability, aligned with the ISO/IEC 25010 software quality standard. Representative items include:

1. All features of the system function correctly and produce expected outputs.
2. The offline detection module operates reliably without internet connectivity.
3. The system responds to user inputs within an acceptable time frame.
4. The GIS mapping and heatmap features accurately reflect location data.
5. The community forum is correctly integrated with the detection module.
6. The overall design and interface are appropriate for the intended user base.

7. The system meets the functional requirements defined in the study's objectives.

The results of both evaluations will be analyzed using weighted mean scores and interpreted against a descriptive scale to determine the overall acceptability and quality of the CORNGUARD system.

3.5.3 User Acceptance Testing (UAT)

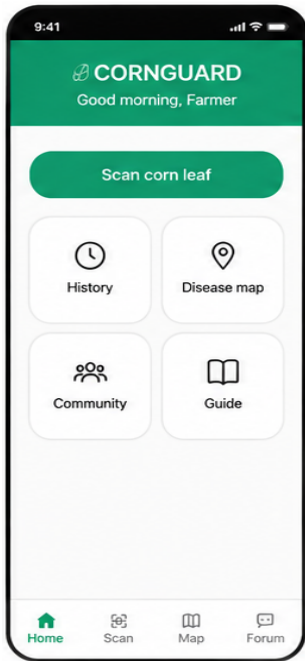
User Acceptance Testing (UAT) will be conducted with a selected group of local corn farmers in Bukidnon to evaluate whether CORNGUARD meets the practical needs of its intended users.

Participants will be recruited from farming communities in the region and will represent the primary target user profile: smallholder farmers with basic smartphone experience. Each participant will be asked to perform a set of predefined tasks — capturing a corn leaf image, reading the diagnosis result, viewing the treatment recommendation, browsing the disease map, and posting a report to the community forum — without assistance from the development team.

Task completion rates, time-on-task, and error counts will be recorded. After completing the tasks, participants will answer a structured questionnaire assessing the application's ease of use, clarity of instructions, usefulness of the diagnosis results, and overall satisfaction.

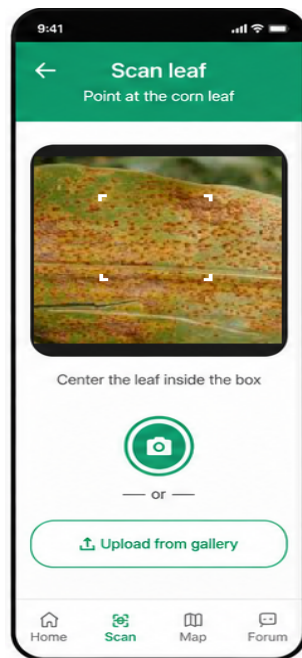
Feedback gathered during UAT will be used to identify usability issues and inform final refinements before deployment

3.5.4 User Interface and Experience Design



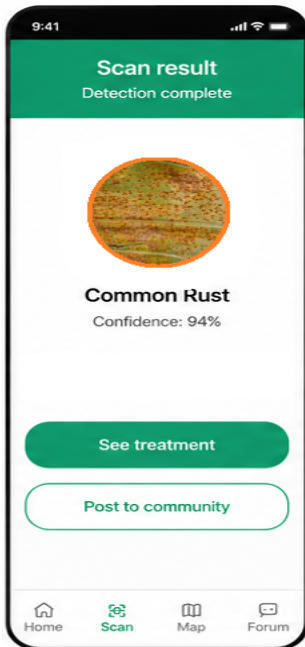
1. Home

1. **Home** Screen. Presents four large icon-driven action buttons: Scan Corn Leaf, Disease History, Disease Map (with heat map indicator), and Community/Notifications. A greeting message addresses the farmer by name.



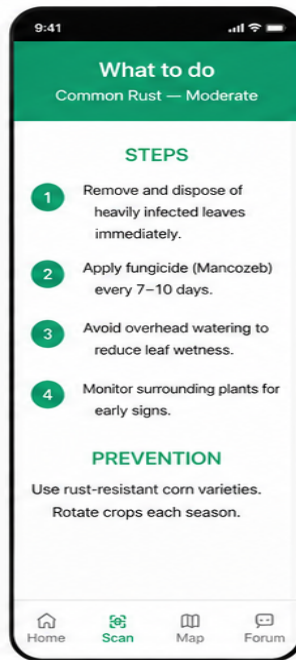
2. Camera

2. **Camera Capture** Screen. Opens the device camera with a rectangular overlay frame instructing the farmer to center the corn leaf. A large circular shutter button and an alternative gallery upload button are provided.



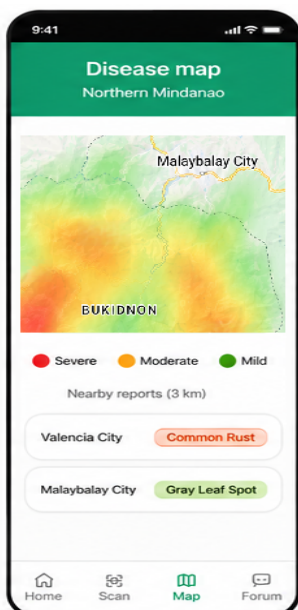
3. Result

3. **Diagnosis Result** Screen. Displays the detected disease name and confidence percentage. Two action buttons allow the farmer to view the treatment recommendation or share the result directly to the community notification feed.



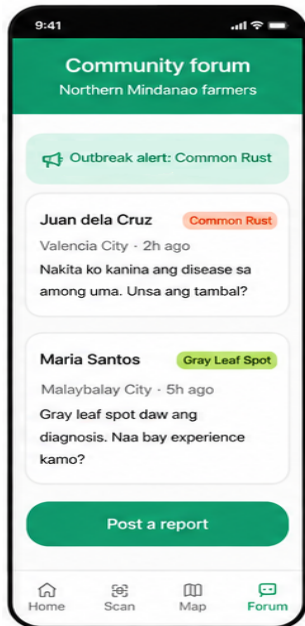
4. Treatment

4. **Treatment Recommendation** Screen. Presents a numbered step-by-step treatment guide and a prevention section, all stored locally and accessible offline.



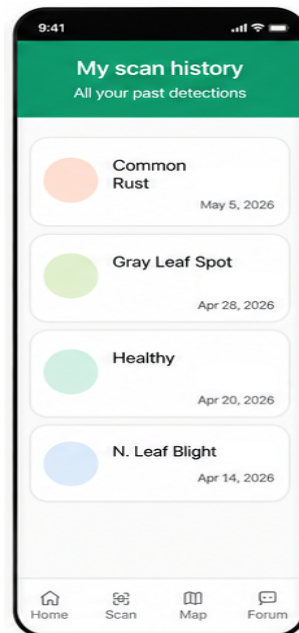
5. Disease map

5. **GIS Disease Map / Heat Map** Screen. Displays an interactive geographic view of Bukidnon with a heat map overlay representing outbreak concentration. The geographic hierarchy (Province → City/Municipality → Barangay → GPS pinpoint) is navigable by zooming into the map. A nearby reports panel lists recent detections within a user-configurable radius.



6. Community

6. **Community Notification / Forum** Screen. Displays a feed of disease detection reports and community posts from farmers in the same municipality or province. A prominent Post a Report button allows farmers to submit reports, optionally pre-populated from a completed scan. Push notification alerts from the system are displayed at the top of this screen.



7. History

7. **Scan History** Screen. A chronological log of all past disease detections performed on the device, stored locally using SQLite and fully accessible offline.

Figure 13. User Interface Wireframes / Screen Mockups.

3.6 Project Timeline

The development of CORNGUARD follows a nine-month timeline from June 2026 to February 2027, structured in accordance with the Agile SDLC framework adopted for this project.

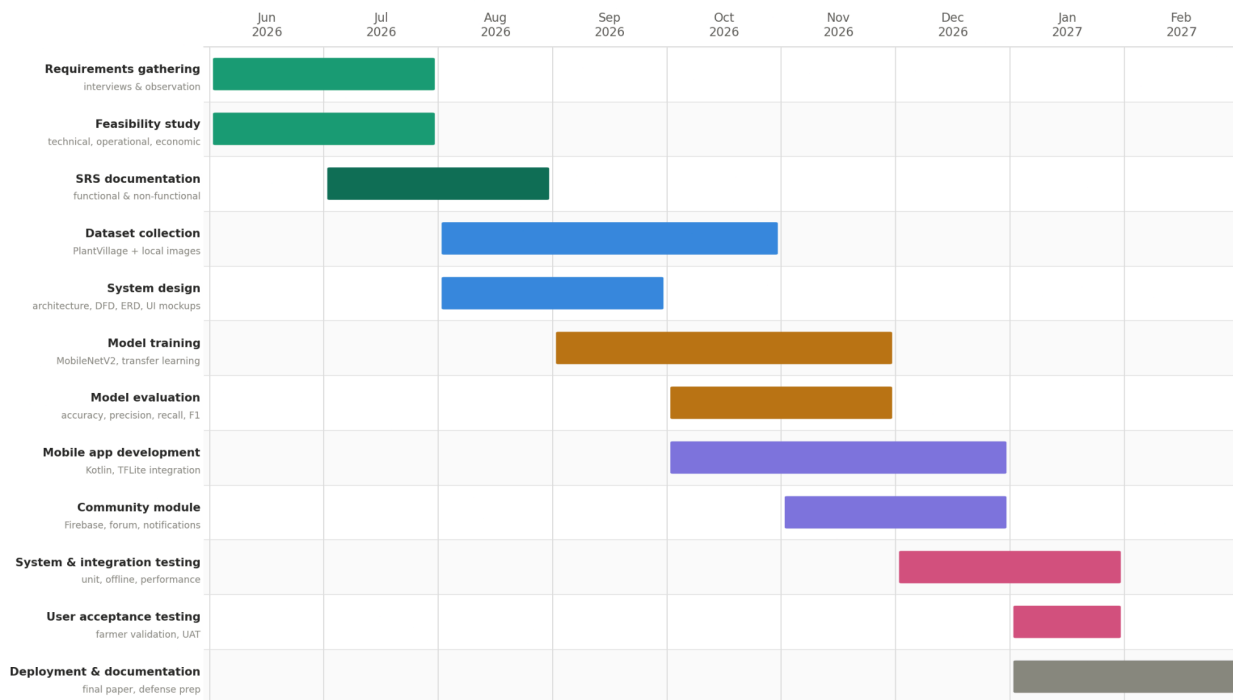


Figure 14: Gantt Chart

The project begins in June and July 2026 with requirements gathering, farmer interviews, feasibility study, and SRS documentation. August through October 2026 focuses on dataset collection from both public repositories and local Bukidnon farms, alongside the completion of all system design outputs including the architecture, DFD, ERD, and UI mockups.

Model training using MobileNetV2 and transfer learning takes place from September to November 2026, with model evaluation running concurrently in October and November to verify that the minimum accuracy threshold is met. Mobile application development runs from October

to December 2026, with the Firebase-powered community module developed in November and December.

System testing and User Acceptance Testing with local farmers are conducted from December 2026 to January 2027. The final two months of January and February 2027 are dedicated to documentation and final paper completion.

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